

**PARTIAL DISCHARGE CLASSIFICATION
USING MACHINE LEARNING**

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PARTIAL DISCHARGE CLASSIFICATION USING MACHINE LEARNING

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PARTIAL DISCHARGE CLASSIFICATION USING MACHINE LEARNING

I MALCOLM SABINUS

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ABSTRACT

Partial discharge (PD) is a critical indicator of insulation degradation in high-voltage electrical equipment, and accurate PD classification is essential for condition-based maintenance and early fault prevention. Conventional PD diagnostic approaches rely heavily on expert interpretation of phase-resolved discharge patterns, which are subjective, time-consuming, and vulnerable to noise and complex operating conditions. To address these limitations, this study investigates the use of traditional machine learning techniques for automated partial discharge classification using a structured, multi-domain feature-based framework. A publicly available partial discharge dataset was utilized, and features were extracted from time-domain, statistical, frequency-domain, and time–frequency representations to capture the physical and spectral characteristics of PD signals. Two independent feature selection strategies—Featurewiz and MLJAR AutoML—were employed to identify the most informative features while reducing dimensionality. Several traditional machine learning classifiers, including Support Vector Machine, Random Forest, Decision Tree, k-Nearest Neighbors, and Artificial Neural Network models, were trained and evaluated using stratified k-fold cross-validation. Model performance was assessed using accuracy, precision, recall, and F1-score metrics. The results show that Random Forest and Artificial Neural Network classifiers consistently achieved the strongest performance across both feature selection strategies, with classification accuracies exceeding 90% and stable F1-scores. Despite differences in feature selection mechanisms, both approaches identified a common core of discriminative features, particularly those related to phase-resolved discharge behavior and energy distribution across frequency bands. Overall, this study confirms the practical viability of traditional machine learning approaches for robust, interpretable, and computationally efficient partial discharge classification in real-world power system monitoring applications.

ABSTRAK

Pelepasan separa (Partial Discharge, PD) merupakan petunjuk awal yang kritikal terhadap degradasi penebatan dalam peralatan elektrik voltan tinggi, dan pengelasan PD yang tepat adalah penting bagi penyelenggaraan berasaskan keadaan serta pencegahan kegagalan awal. Kaedah diagnostik PD konvensional bergantung secara meluas kepada tafsiran pakar terhadap corak pelepasan terlerai fasa, yang bersifat subjektif, memakan masa, dan terdedah kepada gangguan hingar serta keadaan operasi yang kompleks. Bagi mengatasi kekangan ini, kajian ini menilai penggunaan teknik pembelajaran mesin tradisional untuk pengelasan pelepasan separa secara automatik menggunakan rangka kerja ciri berbilang domain yang tersusun. Set data pelepasan separa yang tersedia secara umum telah digunakan, dan ciri-ciri diekstrak daripada domain masa, statistik, domain frekuensi, dan masa–frekuensi bagi menangkap ciri fizikal dan spektrum isyarat PD. Dua strategi pemilihan ciri yang bebas, iaitu Featurewiz dan MLJAR AutoML, telah diaplikasikan bagi mengenal pasti ciri paling diskriminatif serta mengurangkan dimensi ciri. Beberapa pengelas pembelajaran mesin tradisional, termasuk Support Vector Machine, Random Forest, Decision Tree, k-Nearest Neighbors, dan Artificial Neural Network, telah dilatih dan dinilai menggunakan pengesanan silang k-lipatan berstrata. Prestasi model dinilai berdasarkan metrik ketepatan, kejituan, kebolehan panggilan semula, dan skor F1. Keputusan menunjukkan bahawa pengelas Random Forest dan Artificial Neural Network secara konsisten mencapai prestasi terbaik merentas kedua-dua strategi pemilihan ciri, dengan ketepatan pengelasan melebihi 90% dan skor F1 yang stabil. Walaupun terdapat perbezaan dalam mekanisme pemilihan ciri, kedua-dua pendekatan mengenal pasti satu set teras ciri yang signifikan, khususnya yang berkaitan dengan tingkah laku pelepasan terlerai fasa dan taburan tenaga merentasi jalur frekuensi. Secara keseluruhan, kajian ini mengesahkan kebolehlaksanaan praktikal kaedah pembelajaran mesin tradisional untuk pengelasan pelepasan separa yang teguh, boleh ditafsir, dan cekap dari segi pengiraan dalam aplikasi pemantauan sistem kuasa sebenar.

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LIST OF ABBREVIATIONS

AVT	Auditory Vigilance Task
BMM	Bio-Mathematical Models
LLM	Large Language Model
UMS	Universiti Malaysia Sabah
BL	Blended Learning
EEG	Electroencephalography
EOG	Electrooculography
CF	Common Feature
SVM	Support Vector Machine
SWA	Slow-Wave Activity
MLO	Machine Learning Object
NB	Naive Bayes
ANN	Artificial Neural Network
CDR	Cognitive Degradation Rating
PERCLOS	Percentage Eyelid Closure
EM	Electromyography
FE	Feature Extraction
PD	Partial Discharge

CHAPTER 1

INTRODUCTION

1.1 Background

Partial discharge is defined as a localized dielectric breakdown of a small portion of a solid or fluid electrical insulation system under high voltage stress, which does not completely bridge the electrodes (Lu et al., 2020). It typically originates from insulation defects such as voids, cracks, surface contamination, or imperfections within high-voltage electrical equipment (Hussain et al., 2021; Agarwal et al., 2023). These localized discharges degrade the insulation over time, potentially leading to insulation failure and catastrophic system breakdowns (Choudhary et al., 2024). Partial discharge plays a pivotal role as an early warning indicator in various high-voltage components including transformers, cable terminations, switchgear, and Gas Insulated Switchgear (Hussain et al., 2021; Govindarajan et al., 2023). In particular, cross-linked polyethylene cables, widely adopted in countries like Malaysia for power distribution, are vulnerable to partial discharge-induced failures due to aging, improper jointing, or manufacturing defects (Rosle et al., 2021; Choudhary et al., 2024). The consequences of undetected partial discharge in such infrastructure can be severe, ranging from localized blackouts to significant financial losses and service disruption. Reliable detection and classification of partial discharge events are complicated by the transient and stochastic nature of the discharges, as well as the influence of environmental noise, electromagnetic interference, and the presence of multiple overlapping discharge sources (Baug et al., 2017; Ghosh et al., 2020). Traditional diagnostic techniques, such as visual inspection and periodic manual testing, are increasingly being replaced or augmented by automated systems capable of continuous monitoring and intelligent analysis (Sikorski et al., 2020).

The complexity of modern electrical networks and the increasing demand for uninterrupted power supply have driven the need for smarter, more resilient diagnostic frameworks (Vatsa et al., 2024). In this context, artificial intelligence, particularly machine learning and deep learning, have emerged as powerful tools for partial discharge signal interpretation (Lu et al., 2020; Long et al., 2024). These techniques offer capabilities for pattern recognition, noise filtering, and real-time classification, thereby enhancing the accuracy and speed of partial discharge diagnosis (Song et al., 2018; Abdullah et al., 2019). As smart grids and condition-based monitoring systems become the norm, the development of robust, scalable, and noise-resilient partial discharge detection frameworks is imperative (Jia et al., 2020; J. Kim & Kim, 2021). These systems are not only critical to grid reliability and asset longevity but also align with global sustainability goals and digital grid modernization initiatives such as the EU Clean Energy Package and the U.S. Grid Modernization Initiative.

1.2 Significance of the Issue

Early and accurate detection and classification of partial discharge sources are paramount to ensuring the reliability and longevity of power systems (Lu et al., 2020; Kumar et al., 2024). A robust partial discharge monitoring system enables timely intervention and proactive maintenance, thereby preventing costly outages, equipment damage, and safety hazards. In safety-critical industries such as aviation and transportation, where electrical systems are vital, reliable partial discharge diagnosis becomes even more essential for operational efficiency and risk mitigation (Y. Wang et al., 2023; Jiang et al., 2024).

1.2.1 Local Perspective

From an economic perspective, partial discharge detection and monitoring play a crucial role in reducing maintenance costs and preventing unexpected failures (Sarkar et al., 2021). The global partial discharge Monitoring Systems market was valued at \$529.91 million in 2023 and is projected to reach \$744.01 million by 2029, reflecting the growing awareness of partial discharge monitoring's importance in the electrical industry. The increasing demand for partial discharge monitoring systems in rapidly industrializing regions such as China and India highlights the necessity of reliable power infrastructure for economic growth (Song et al., 2018). Furthermore, government policies focusing on sustainable power generation and grid reliability emphasize the adoption of advanced monitoring techniques, including partial discharge detection, to enhance electrical infrastructure resilience (Abdullah et al., 2019).

1.2.2 Market Size and Investment Trends in partial discharge Monitoring

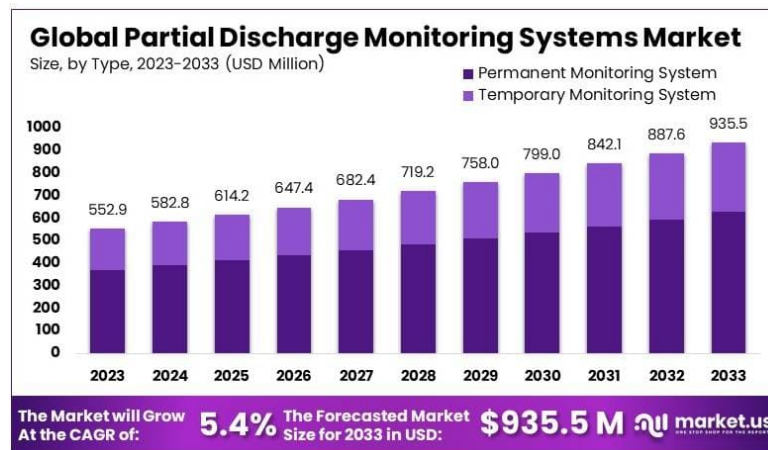


Figure 1.1 Global Partial Discharge Monitoring Systems Market By Type (2023–2033), Showing Growth From USD 552.9 Million In 2023 To USD 935.5 Million By 2033 (Market.us, 2024).

The global market for partial discharge monitoring systems is growing robustly, driven by the need for grid reliability and aging infrastructure. In 2023, the partial discharge monitoring market was valued around USD 550–580 million, and is projected

to reach approximately USD 740–935 million over the next 5–10 years. This reflects a healthy compound annual growth rate on the order of 5–6%. Investments in smart grids and preventive maintenance are key factors propelling this growth (Jia et al., 2020; J. Kim & Park, 2020). Notably, Asia Pacific is the leading region in partial discharge monitoring adoption, as countries like China, India, and Japan heavily invest in such systems to support their large power networks (Song et al., 2018; Abdullah et al., 2019). North America and Europe also represent significant markets, with North America comprising roughly one-third of demand (about 35% in 2023) amid major grid modernization efforts. Malaysia, as part of the dynamic Asia Pacific region, is seeing increased attention to partial discharge monitoring through its national utility initiatives, although its domestic market size is a modest portion of the global figure (Chan et al., 2023). The trend worldwide is clear: utilities and industries are allocating more budget to partial discharge detection technology as part of broader condition-based maintenance and grid reliability programs (Sikorski et al., 2020).

From an investment standpoint, global utilities and governments are channeling substantial funds into grid monitoring technologies, including partial discharge systems. For example, the European Union has mandated regular partial discharge assessments and earmarked about €200 million specifically for upgrading infrastructure with advanced partial discharge monitoring by 2025. In the United States, utilities invested roughly USD 168 billion into the electric grid in 2023, with about USD 30.7 billion directed to transmission and USD 57.1 billion to distribution upgrades. A portion of these investments goes toward installing sensors and online monitoring (such as partial discharge detectors) to preempt failures (Liccardo et al., 2022). Government-backed programs, such as the U.S. Department of Energy’s Grid Resilience and Innovation Partnerships, totaling USD 2.7 billion in funding, further illustrate the strong emphasis on modernizing grid monitoring and diagnostic capabilities. In Asia, state-owned utilities are adopting similar approaches for example, State Grid China has deployed innovative solutions, including UAV-based ultrasonic partial discharge detection systems, to patrol and maintain its extensive transmission network (J. Kim & Park, 2020). Overall, both private-sector investments and public-sector funding mechanisms demonstrate a clear and sustained

trend toward strengthening partial discharge monitoring as a core component of critical infrastructure development.

1.2.3 Economic Impact of partial discharge-Related Outages and Failures

Partial discharge is not just a technical issue; its consequences carry heavy financial costs. Partial discharge activity deteriorates insulation over time and is a leading cause of electrical equipment failures, which in turn can trigger power outages (Hussain et al., 2021; Agarwal et al., 2023). Studies indicate that a substantial proportion of disruptive power apparatus failures—on the order of 80–85% in high-voltage substations—are attributable to partial discharge activity (Sikorski et al., 2020). When such failures occur, the associated economic consequences can be severe, resulting in industrial downtime, revenue losses for utilities, and substantial recovery and repair costs. To put this in perspective, power outages cost businesses and economies billions annually. In the United States, for example, sudden power outages cost businesses over \$150 billion each year. In Malaysia, a single major blackout (the 2003 nationwide outage) was estimated to have cost local businesses about USD 13.8 million.

Even smaller-scale partial discharge incidents can incur tens of thousands of dollars in losses per day for the affected facility. A case study from a wind farm showed that partial discharge-induced tripping of a circuit breaker caused over \$24,000 in lost revenue in just one occurrence. For utilities, a damaged transformer or switchgear that takes down a substation can mean costs reaching hundreds of thousands of dollars per day in unmet demand and repair efforts (Aslam et al., 2022). These direct costs are compounded by indirect losses: lost productivity, spoiled products, and cascading effects on other businesses and services. It's no surprise, then, that avoiding unplanned outages is a top financial priority for power providers and consumers alike. Beyond individual incidents, chronic reliability issues have macroeconomic impacts. Frequent outages erode investor confidence and can stunt economic growth. South Asian

economies, for instance, have been shown to suffer substantial GDP deficits due to power supply disruptions. In the corporate world, downtime can affect everything from manufacturing output to data center operations and safety systems. U.S. industry experts argue that many outage cost estimates actually understate the true impact, since they often fail to capture follow-on losses across supply chains and society. In summary, partial discharge related failures impose substantial economic costs, thereby underscoring the importance of investing in partial discharge monitoring systems, as the cost of preventive measures is significantly lower than the cost associated with unanticipated equipment failure.

1.2.4 Government Funding and Policies on partial discharge Monitoring Infrastructure

Governments and regulators worldwide recognize the importance of grid reliability and have introduced policies and funding initiatives that, directly or indirectly, support partial discharge monitoring deployment. Malaysia's government, through the Energy Commission (Suruhanjaya Tenaga) and regulatory frameworks such as Incentive-Based Regulation, has established system reliability as a key performance metric for electric utilities. Under the Incentive-Based Regulation framework, Tenaga Nasional Berhad—Malaysia's primary electric utility—is incentivized to invest in infrastructure upgrades and preventive maintenance strategies aimed at minimizing supply interruptions. This policy approach has resulted in sustained investments in technologies such as online partial discharge detection systems, high-speed protection schemes, and systematic asset replacement programs to enhance grid resilience (Abdullah et al., 2019). The effectiveness of these measures is reflected in the System Average Interruption Duration Index for Peninsular Malaysia, which was recorded at only 48.13 minutes per customer in 2019, outperforming many developed economies. This level of reliability is attributed in part to the proactive identification and mitigation of incipient insulation faults, including partial discharge activity, before they escalate into major system

failures. The Malaysian government also supports research and development in grid monitoring through agencies like Tenaga Nasional Berhad Research and various university grants, fostering local expertise in partial discharge detection techniques (Chan et al., 2023).

Elsewhere in Asia, governments are bolstering grid resilience which includes partial discharge monitoring as a component. China's government, via State Grid and Southern Grid, has poured funding into modernizing its vast transmission network – including installing continuous monitoring sensors and even using drones for partial discharge inspections (J. Kim & Park, 2020). India has launched programs to upgrade its aging transmission and distribution infrastructure, backed by government financing and multilateral loans, which often stipulate the adoption of better monitoring to reduce technical losses and outages. Southeast Asian countries like Singapore and Thailand have also updated their grid codes and standards to encourage condition-based maintenance; utilities in these countries increasingly use partial discharge testing as part of equipment commissioning and routine checks (D. Wang et al., 2024). On the international stage, organizations such as the International Electrotechnical Commission have set standards (e.g. IEC 60270) that many governments adopt or reference, effectively making partial discharge measurement a normative requirement during factory testing and on-site commissioning of high-voltage equipment (Vera et al., 2023). Additionally, the World Bank and regional development banks fund electrification and grid reliability projects in developing nations, often including components for training and equipment to implement partial discharge monitoring as a means to extend asset life.

In advanced economies, regulatory pressure is perhaps most formalized in Europe and North America. Several European countries mandate periodic partial discharge testing for critical grid assets as part of their preventive maintenance regulations. The EU's focus on smart grids and its large funding programs (like the European Green Deal investments in energy networks) explicitly list grid diagnostic technologies eligible for support. In the United States, while no federal law specifically compels partial discharge monitoring, the trend is guided by reliability standards and incentives. The

North American Electric Reliability Corporation has reliability standards that push utilities toward robust maintenance of key equipment (transformers, breakers, etc.), and partial discharge monitoring is recommended as a best practice in industry guidelines for preventing insulation failures (Hussain et al., 2021). U.S. federal funding through the Department of Energy's grid modernization initiatives and infrastructure bills has provided billions for grid upgrades, implicitly encouraging utilities to include advanced sensors (partial discharge, thermal, etc.) in their plans. In summary, government policies worldwide increasingly favor proactive maintenance and infrastructure resiliency, creating an environment where partial discharge monitoring is not just optional but often a supported or expected investment. This alignment of policy and funding with the goal of reducing blackouts has accelerated the adoption of partial discharge monitoring technology on a global scale.

1.2.5 Industry Initiatives and partial discharge Monitoring Programs

Electric utilities and industry players are at the forefront of implementing partial discharge monitoring as part of their asset management strategies. In Malaysia, both Tenaga Nasional Berhad and Sabah Electricity Sdn. Bhd. have launched initiatives to improve grid reliability that include partial discharge management. Tenaga Nasional Berhad, which serves Peninsular Malaysia, has been a regional leader in reliability and has systematically rolled out condition-based maintenance programs (Abdullah et al., 2019). As part of these efforts, Tenaga Nasional Berhad pioneered partial discharge mapping techniques on its medium-voltage cables, conducting extensive field trials as early as the 2000s to evaluate their effectiveness. Today, Tenaga Nasional Berhad employs online partial discharge detection systems and partial discharge mapping in its operations for example, to continuously monitor underground power cables and detect insulation defects before they escalate (Abdullah et al., 2019). According to Tenaga Nasional Berhad's recent reports, integrating partial discharge monitoring has been instrumental in early fault identification and has helped mitigate the likelihood of

major failures in the distribution network. Tenaga Nasional Berhad has extended these predictive maintenance practices to critical equipment like 33 kV switchgears, again using partial discharge sensing to preempt flashovers and catastrophic breakdowns (Alsumaidae et al., 2022). By catching warning signs like partial discharge, Tenaga Nasional Berhad can schedule maintenance or component replacements proactively, thereby keeping its System Average Interruption Duration Index and System Average Interruption Frequency Index indices at impressively low levels. This approach directly translates to fewer outages and improved service continuity for customers.

Sabah Electricity Sdn. Bhd., which manages the grid in Sabah, has also recognized the importance of such monitoring, albeit starting from a more challenging reliability baseline. Historically, Sabah experienced frequent power supply disruptions; in 2005, the System Average Interruption Duration Index exceeded 4,000 minutes per customer, reflecting a very low level of system reliability. Through various programs, Sabah Electricity Sdn. Bhd. has dramatically improved this, reducing System Average Interruption Duration Index to about 274 minutes in 2023 and setting a target of 100 minutes by 2030. To achieve this, Sabah Electricity Sdn. Bhd. is upgrading its infrastructure and maintenance regimes. While specific partial discharge monitoring programs at Sabah Electricity Sdn. Bhd. are less publicized than Tenaga Nasional Berhad's, the utility is known to be collaborating with Tenaga Nasional Berhad and technical partners to adopt similar best practices. This likely includes using portable partial discharge detection units for surveying substations and cables and prioritizing the replacement of equipment that shows partial discharge symptoms (Sikorski & Gielniak, 2025). The Malaysian government has been supportive, with federal funds aiding Sabah's grid improvements (including a subsidy to cover Sabah Electricity Sdn. Bhd.'s operational deficits while upgrades are underway). Industry-wide, there is also knowledge sharing: Tenaga Nasional Berhad Research and local universities have worked on partial discharge detection techniques suited for the tropical environment (where high humidity can exacerbate surface discharge) (Azam et al., 2023). These RD outputs benefit Sabah Electricity Sdn. Bhd. and other regional utilities in crafting their partial discharge monitoring and mitigation strategies.

Beyond the utilities, industry involvement includes equipment manufacturers and service providers who run partial discharge monitoring programs or services. Companies like Doble, Megger, Qualitrol, and EA Technology offer partial discharge testing as part of substation maintenance contracts and have conducted training workshops in Malaysia and across Asia to build local capacity. Some large industrial consumers (e.g. semiconductor factories or oil gas facilities in Malaysia) have also started installing partial discharge monitors on their critical high-voltage equipment to avoid costly downtime. For instance, continuous partial discharge monitoring systems are now available for large motors, generators, and gas-insulated switchgear and vendors often cite success stories where their systems gave early warnings that averted a failure (Seri et al., 2021). An example of such detection technology is illustrated in Figure 1.2. Global industry statistics further underscore the value of partial discharge monitoring; one industry source reports that approximately 85% of disruptive substation failures are attributable to partial discharge activity, indicating that the implementation of partial discharge monitoring programs can significantly reduce the risk of such failures (Sikorski et al., 2020). Another industry report highlighted that insulation failure (often traceable to partial discharge) contributes to up to 90% of switchgear defects, reinforcing why so many maintenance teams now include partial discharge checks as a standard procedure (Alsumaidae et al., 2022).

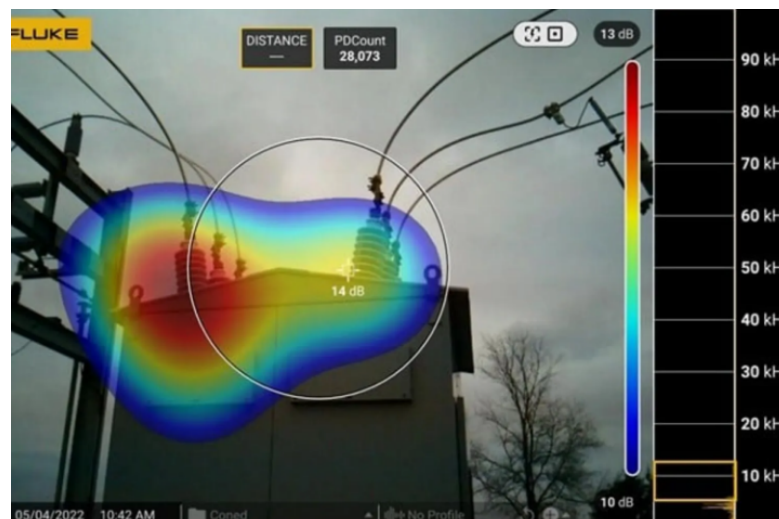


Figure 1.2 Example of partial discharge detection using an acoustic imaging camera (Fluke ii910), where colored contours indicate partial discharge intensity on substation bushings. (Fluke Corporation, n.d.)

In the Malaysian context, the collaboration between industry and the national utility is evident in programs like the Intelligent Predictive & Diagnostic Monitoring system mentioned by Tenaga Nasional Berhad, which integrates partial discharge sensors in the broader smart grid monitoring platform. Tenaga Nasional Berhad and Sabah Electricity Sdn. Bhd. also participate in regional forums (through ASEAN utilities and international bodies) to share incident statistics and lessons. While detailed partial discharge incident statistics in Malaysia are not publicly released, anecdotal evidence and case reports suggest that numerous minor faults have been caught and resolved thanks to partial discharge detection, preventing them from becoming major outages. Each avoided incident contributes to improved reliability metrics. The drop in Tenaga Nasional Berhad's transmission system minutes (only 0.27 minutes of interruption in 2019 for Peninsular Malaysia) is partly credited to such preventive monitoring. Regionally, countries like Singapore report very low outage rates as well, attributable to stringent monitoring and maintenance regimens (which include partial discharge testing). This industry-wide push from utility engineers patrolling with ultrasonic partial discharge detectors, to control centers analyzing continuous online partial discharge data highlights a robust commitment to nipping problems in the bud (Abdullah et al., 2019; Alshalawi & Al-Ismail, 2024).

1.3 Existing Approaches and Challenges

The diagnosis of partial discharge (PD) is a critical aspect of condition monitoring for high-voltage equipment, providing essential insights into the health of insulation systems (Hussain et al., 2021). Diagnostic approaches have evolved from traditional methods, which rely on the manual interpretation of discharge patterns, to modern automated systems employing artificial intelligence and machine learning (Mas'ud et al., 2016). While conventional techniques have been foundational, they are often limited by subjectivity, inefficiency in handling large datasets, and susceptibility to noise, which can compromise diagnostic accuracy (Florkowski, 2021a; Das et al., 2023).

To address this issue, automated methods have been developed to provide more objective and efficient analysis (Lu et al., 2020). However, these advanced systems introduce their own set of challenges, including the complexity of robust feature extraction, the need for effective noise cancellation, and ensuring model generalization across varied operational conditions (Friebe & Jenau, 2024; Long et al., 2024).

1.3.1 Limitations of Traditional Methods

Conventional methods for diagnosing partial discharge phenomena primarily rely on the manual interpretation of Phase-Resolved partial discharge patterns by skilled operators (Kasinathan et al., 2017; Florkowski, 2021a). This approach, while historically effective, is increasingly recognized for its inherent limitations, which hinder its applicability in modern high-voltage insulation systems.

One of the foremost challenges in manual partial discharge analysis is its subjectivity. The interpretation of Phase-Resolved partial discharge patterns depends heavily on the experience and expertise of the operator, making it susceptible to human error (Mas'ud et al., 2016; Lu et al., 2020). Variations in individual judgment can lead to inconsistencies in diagnosis, ultimately affecting the reliability of the results (Kunicki et al., 2018). Such subjectivity poses a significant risk in critical applications where precise partial discharge identification is essential for ensuring equipment longevity and operational safety.

Another major drawback is the time-consuming nature of manual partial discharge analysis. Evaluating large volumes of Phase-Resolved partial discharge data requires substantial effort and expertise, making the process inefficient, particularly for large-scale industrial applications (Long et al., 2024). As the demand for faster and more efficient partial discharge diagnosis grows, the reliance on human interpretation becomes increasingly impractical, emphasizing the need for automated techniques that

can process large datasets more efficiently (Das et al., 2023).

Furthermore, traditional partial discharge diagnosis methods are highly susceptible to noise interference, which poses a significant challenge in real-world applications (Friebe & Jenau, 2024). In practical scenarios, especially in on-site measurements, partial discharge signals are often masked by background noise, making it difficult to distinguish genuine partial discharge activity from external disturbances (Ghosh et al., 2020; J. Bao et al., 2024). Conventional filtering techniques are not always effective in isolating partial discharge signals, leading to potential misclassification or overlooked defects (Hussein et al., 2016). The presence of noise-induced errors further complicates the reliability of manual Phase-Resolved partial discharge pattern interpretation.

In addition to noise susceptibility, conventional techniques struggle to handle complex partial discharge scenarios where multiple partial discharge sources coexist within an insulation system (Baug et al., 2017). In such cases, overlapping partial discharge patterns make it challenging to extract meaningful features and perform accurate classification (Biswas et al., 2016; Firuzi et al., 2017). Traditional methods often lack the capability to effectively differentiate between multiple partial discharge sources, limiting their diagnostic accuracy and effectiveness. This limitation is particularly problematic in aging power equipment, where different types of insulation defects may simultaneously contribute to partial discharge activity.

1.3.2 Challenges with Automated Methods

Given these challenges discussed in Section 1.3.1, there is a growing need for advanced partial discharge diagnosis techniques that can overcome the limitations of manual interpretation. The integration of machine learning and signal processing methods has shown promising potential in addressing these issues by providing objective, efficient, and noise-resilient partial discharge classification (Mas'ud et al., 2016; Lu et al.,

2020). These approaches offer significant advantages, including improved efficiency, objectivity, and the ability to process large volumes of data. However, despite their potential, automated methods still face several fundamental challenges that must be addressed to achieve reliable and accurate partial discharge classification.

One of the primary challenges in automated partial discharge classification is the process of feature extraction. Effective partial discharge analysis requires the selection of meaningful features that can accurately represent different types of discharge activity (Long et al., 2024). However, partial discharge signals are complex and often exhibit highly stochastic behavior, making it difficult to determine which features are most relevant for classification (Dai et al., 2017). The process of selecting and engineering features typically requires significant domain expertise to ensure that extracted parameters adequately capture the characteristics of different partial discharge types (W. Zhou et al., 2017). Without careful feature selection, machine learning models may struggle to distinguish between genuine partial discharge signals and external noise, leading to misclassification and unreliable diagnostic outcomes.

Another significant obstacle is the optimization of machine learning algorithms for partial discharge classification. Selecting the most suitable algorithm is not always straightforward, as different techniques exhibit varying levels of effectiveness depending on the nature of the partial discharge data (Fei et al., 2024b). The stochastic nature of partial discharge signals further complicates this process, as the variability in signal characteristics makes it difficult to establish a universally optimal model (Vigneshwaran et al., 2022). Furthermore, machine learning algorithms often require fine-tuning of hyperparameters to achieve high classification accuracy. This optimization process can be computationally intensive and may necessitate extensive trial-and-error experimentation to identify the best-performing configurations.

Beyond algorithm selection and optimization, a major concern with automated partial discharge classification is the issue of generalization. Many machine learning models struggle to generalize well when applied to new or unseen data (Raymond et al., 2022).

This problem is particularly pronounced when models are trained on limited or imbalanced datasets, where certain partial discharge types may be underrepresented (V.-N. Tuyet-Doan et al., 2021). As a result, the trained models may exhibit high accuracy on the training data but fail to maintain their performance when applied to real-world scenarios. The challenge of generalization necessitates the use of robust data augmentation techniques, transfer learning, or adaptive learning mechanisms to improve model resilience across different operating conditions.

Additionally, the presence of environmental noise and interference remains a significant challenge for automated partial discharge classification systems (Azam et al., 2023). In practical applications, partial discharge signals are often contaminated by external noise sources, making it difficult to isolate and identify discharge patterns accurately (Friebe & Jenau, 2024). The effectiveness of an automated classification system heavily depends on the robustness of its feature extraction and noise filtering techniques. Advanced signal processing methods, such as wavelet transforms, adaptive filtering, and deep learning-based denoising techniques, have been explored to mitigate the impact of noise (Abdullah et al., 2019; T. Jin et al., 2019). However, ensuring the reliability of partial discharge classification in noisy environments remains an active area of research.

1.4 Problem Formulation

1.4.1 Research Gap

Current research in partial discharge detection for overhead power lines often focuses on specific datasets or individual algorithms, lacking a comprehensive comparison of traditional and modern machine learning techniques across diverse datasets (Basharan et al., 2018; Song et al., 2018). This limitation restricts the understanding of the relative strengths and weaknesses of different algorithms under varying conditions, such

as noise levels, environmental factors, and signal acquisition setups. Performance reported on a single dataset may not generalize well to other datasets due to differences in partial discharge signal characteristics, noise profiles, and measurement protocols, necessitating a broader comparison across multiple datasets to assess the robustness and generalizability of different machine learning techniques. The optimal selection and combination of features for partial discharge detection also remain underexplored, with existing studies rarely investigating the impact of different feature sets, such as time-domain, frequency-domain, and deep-learned embeddings, on the performance of various machine learning models (Long et al., 2024). Different algorithms may be sensitive to specific feature types, and optimizing feature selection can significantly enhance detection accuracy and efficiency, making a systematic investigation of feature combinations crucial for identifying the most informative and discriminative features for each algorithm. This gap hinders the development of robust and generalizable partial discharge detection systems that can perform reliably across diverse overhead power line configurations and environmental conditions, and the lack of comprehensive algorithm comparisons and feature analyses makes it challenging to select the most appropriate algorithm and feature set for specific applications. This study aims to address this gap by providing a comparative analysis of traditional and modern machine learning methods across multiple datasets and investigating the impact of different feature combinations on their performance.

1.4.2 Problem Statement

Based on the identified research gap, this study addresses several key problems. First, there is a lack of a comprehensive comparison of machine learning algorithms for partial discharge detection, which includes evaluating the performance of traditional algorithms (Support Vector Machine, Random Forest, K-Nearest Neighbors, Naive Bayes, AdaBoost) in terms of accuracy, precision, recall, F1-score, Area Under the Curve, and computational cost, as well as understanding the trade-offs between performance and

computational complexity, which is critical for real-time partial discharge monitoring in overhead power lines (Lu et al., 2020; Kumar et al., 2024). Second, there is limited understanding of the optimal feature sets for different machine learning algorithms in partial discharge detection, requiring an investigation of various feature extraction techniques, including time-domain features (e.g., pulse amplitude, width), frequency-domain features (e.g., Fast Fourier Transform, wavelet transforms), and deep-learned embeddings, to evaluate their impact on the performance of each algorithm and identify the most discriminative features that contribute significantly to accurate partial discharge classification (Long et al., 2024). Third, there is a need to evaluate the generalizability of partial discharge detection models across diverse datasets, which involves testing trained models on datasets not used during training, representing different noise conditions, environmental factors, and signal acquisition protocols, to provide insights into the robustness and real-world applicability of the developed models for overhead power line monitoring (Raymond et al., 2022).

1.5 Study Objectives

Based on the hypotheses, this study aims to achieve the following objectives:

- To identify the optimal feature set for each machine learning algorithm by evaluating different feature combinations.
- To compare the performance of various traditional machine learning algorithms (Support Vector Machine, Decision Tree, Random Forest, K-Nearest Neighbors, ANN) for partial discharge detection.

1.6 Scope Of Work

The scope of this thesis includes:

- Preprocessing and feature extraction from publicly available partial discharge datasets. This involves applying signal conditioning techniques to remove noise and irrelevant components, followed by the extraction of features such as wavelet coefficients, phase-resolved partial discharge patterns, statistical descriptors, and entropy measures. Feature selection will be guided by existing literature and optimized to capture the characteristics of partial discharge activity under different conditions.
- Training and evaluation of traditional machine learning models using cross-validation on individual datasets and a combined dataset. This includes training each selected machine learning algorithm on individual partial discharge datasets using k-fold cross-validation to ensure statistical robustness. The trained models will then be evaluated on separate test sets as well as on the aggregated dataset to measure generalization performance.
- Comparison of model performance based on accuracy, precision, recall, F1-score, and computational time. These performance metrics will allow for a comprehensive evaluation of each algorithm, accounting for both predictive quality and computational efficiency, which are critical factors in practical deployment.
- Analysis of the impact of different feature combinations on the performance of each model. This involves systematic evaluation of various feature subsets and quantifying their influence on the classification results. The goal is to identify optimal feature sets that maximize detection accuracy while minimizing computational costs, thus supporting real-time monitoring applications.

1.7 Organisation Of Thesis

This thesis is organized as follows: Chapter 1 introduces the background, motivation, problem statement, hypotheses, objectives, and scope of the research. It provides a comprehensive overview of the context, rationale, and goals of the study, setting the stage for the subsequent chapters.

Chapter 2 reviews the existing literature on partial discharge detection, covering various methods and challenges. This chapter provides a detailed analysis of prior research in partial discharge analysis, including traditional machine learning techniques, signal processing approaches, and recent developments in automated diagnostics. It identifies current gaps and motivates the research direction of this study.

Chapter 3 details the methodology employed, including dataset acquisition, signal preprocessing, feature extraction, model selection, and performance evaluation. This chapter outlines the entire experimental pipeline, ensuring clarity and reproducibility. It describes the datasets used, the preprocessing strategies applied, the features extracted from partial discharge signals, and the machine learning algorithms selected for evaluation.

Chapter 4 presents the experimental results, comparing the performance of various machine learning algorithms and analyzing the effects of feature selection. This chapter includes comparative results across multiple datasets, evaluates generalization, and interprets model behavior in relation to feature sets.

Chapter 5 concludes the study by summarizing the key findings, discussing their implications for real-world partial discharge monitoring, and suggesting directions for future research. It highlights the contributions of the work and outlines opportunities to extend the study, such as integration with Internet of Things sensors or hybrid deep learning models.

The Appendix includes supplementary materials such as dataset details, signal transformation formulas, algorithmic configurations, and extended experimental results. This section serves as a technical reference for readers and researchers interested in replicating or building upon this work.

CHAPTER 2

LITERATURE REVIEW

2.1 Machine Learning in partial discharge Detection

The application of machine learning (ML) techniques to partial discharge (PD) analysis has become increasingly prevalent, driven by the need to overcome the limitations inherent in conventional diagnostic methods (Mas'ud et al., 2016; Lu et al., 2020). Historically, PD assessment relied heavily on the manual interpretation of measurement data, such as Phase-Resolved partial discharge (PRPD) patterns, by human experts. This process is often subjective, time-consuming, and can be unreliable, particularly when dealing with the large volumes of data generated by modern online monitoring systems (Seri et al., 2021; Alsumidaee et al., 2022). Traditional approaches also face significant challenges in noisy operational environments, where distinguishing PD signals from external interference and background noise is a primary obstacle (Ghosh et al., 2020; Park et al., 2023). Moreover, identifying and separating signals from multiple concurrent PD sources (Baug et al., 2017; Firuzi et al., 2017) and reliably quantifying the severity or danger of an insulation defect remain complex tasks for conventional analysis (Ma et al., 2021; Orellana et al., 2023). Machine learning offers a pathway to automated, objective, and more accurate PD analysis, forming the core of advanced condition monitoring and diagnostic systems (Lu et al., 2020). A primary application of ML in this field is the classification of PD source type such as differentiating between corona, surface discharges, and internal voids which is essential for assessing the risk posed to insulation integrity (Anjum et al., 2017; Borghei & Ghassemi, 2022). Another key task is anomaly detection, where ML models are trained to identify PD events as deviations from normal operational signals or ambient noise, a capability crucial for robust

online monitoring (Thi et al., 2020; Florkowski, 2021b). Furthermore, ML algorithms are increasingly used for the localization of PD sources, estimating their physical position within high-voltage (HV) equipment, which facilitates targeted maintenance and repair (Liu et al., 2020; Chan et al., 2023). The overarching motivation for employing ML is to develop intelligent systems that can learn complex patterns directly from sensor data, adapt to changing operational conditions, and provide consistent and reliable diagnostic assessments (Sikorski & Gielniak, 2025).

2.2 Overview of Machine Learning Algorithms for partial discharge Detection

A variety of traditional machine learning algorithms have been employed for partial discharge (PD) detection and analysis. These algorithms are primarily used for tasks such as the classification of PD types, localization of PD sources, and noise rejection. This section provides an overview of some commonly used traditional ML algorithms found in the reviewed literature.

2.2.1 Support Vector Machines (SVM)

Support Vector Machines (SVMs) are supervised learning models that analyze data for classification and regression. In classification tasks, SVMs construct an optimal hyperplane in a high-dimensional space to serve as a decision boundary with the largest possible margin to the nearest data points of any class. SVMs have been widely used for classifying PD sources and defect types. For instance, SVM has been effectively utilized to recognize PDs in switchgear from ultra-high frequency (UHF) signal data (H. Jin et al., 2018) and to classify four different PD types using an optimized model (Dai et al., 2017). They are also used for localizing PD sources in GIS based on image edge detection features (X. Li et al., 2019). Advanced applications include using a Brain

Storm Optimization (BSO) enhanced SVM for pattern recognition of ultrasonic PD signals in transformers (S. Zhou et al., 2022). The selection of SVM is often motivated by its strong theoretical foundation, effectiveness in high-dimensional spaces, and robust performance even with limited training data (W. Zhou et al., 2017; Bin et al., 2020). Its ability to handle the complex characteristics of PD data and its strong performance in multi-class classification make it a preferred choice for pattern recognition and distinguishing between different PD types (H. Jin et al., 2018). Furthermore, One-Class SVM (OCSVM) has been chosen for its ability to model normal, PD-free conditions without requiring a fully labeled training set, which is advantageous for online monitoring systems operating in low signal-to-noise ratio (SNR) environments. For defect classification in ceramic insulators, SVM outperformed k-NN, Naive Bayes, and Decision Tree with 90.6% accuracy (Ma et al., 2021). For aging state recognition in polyethylene-insulated cables, SVM achieved up to 100% accuracy with a small subset of features (Morette et al., 2020).

2.2.2 Random Forest (RF)

Random Forest (RF) is an ensemble learning method that constructs a multitude of decision trees during training. It outputs the class selected by the most trees for classification tasks, making it highly regarded for its robustness against overfitting and its ability to manage high-dimensional data. RF has been successfully applied in various PD detection scenarios, including identifying the location of PD sources (Bag et al., 2019) and classifying PD types (Prabhu et al., 2016; Guo et al., 2021). Specific applications include locating PD sources using features from optical sensor data (Bag et al., 2019), identifying faults based on the concentration of decomposition gases (Tan et al., 2022), and classifying defects in on-board cable terminals for high-speed railways (Guo et al., 2021). The choice of RF is often driven by its high accuracy, its ability to process large datasets with numerous input variables, and its built-in feature ranking capabilities (Bag et al., 2019). In terms of performance, RF has consistently demonstrated high accuracy.

A PD source location accuracy of 95.6% was achieved using RF on optical sensor data (Bag et al., 2019).

2.2.3 Decision Trees (DT)

Decision Trees are a non-parametric supervised learning method used for classification and regression. They build a model that predicts a target variable's value by learning simple, interpretable decision rules inferred from data features. Decision trees have been applied for PD fault type recognition (Lee, 2020) and for detecting insulation faults in transformers, where they have shown high accuracy for specific fault classes (Maseko et al., 2025). These models are often chosen for their interpretability and their capacity to handle both numerical and categorical data. They have also served as a baseline model for comparison against more complex algorithms like Back Propagation Neural Networks (BPNN) (Fei et al., 2024a). In terms of performance, a Fuzzy Logic Clustering Decision Tree (FLCDT) was shown to perform significantly better than standard decision tree algorithms like CART and See5 in terms of classification precision for transformer diagnostics (Lee, 2020). Furthermore, an accuracy of 100% was achieved for specific insulation faults, such as those caused by thermal aging, using (Maseko et al., 2025).

2.2.4 k-Nearest Neighbors (k-NN)

k-Nearest Neighbors (k-NN) is a simple, instance-based learning algorithm. For classification, an object is assigned to the class most common among its k nearest neighbors, as determined by a majority vote. k-NN has been applied to classify PD types in GIS (Bin et al., 2020; Fei et al., 2024a) and for online stator winding fault diagnosis in induction motors (Sarkar et al., 2021). The algorithm is frequently chosen

for its simplicity, effectiveness, and ease of implementation, especially in scenarios with irregular decision boundaries (Bin et al., 2020). It has also been noted for enabling effective online fault detection with high prediction accuracy for motor faults (Sarkar et al., 2021). In terms of performance, k-NN achieved an accuracy of 88.5% in classifying four types of PD in GIS (Bin et al., 2020). In motor fault diagnosis, it demonstrated acceptable prediction accuracy, making it viable for online monitoring systems (Sarkar et al., 2021). While serving as a baseline model, k-NN's results have been compared against ANNs that achieved classification accuracies as high as 96.03% (Haiba & Eliwa Gad, 2024).

2.2.5 Artificial Neural Networks (ANN) (Traditional Architectures)

Artificial Neural Networks (ANNs), in their traditional forms like multilayer perceptrons (MLPs) trained with back-propagation, are computational models inspired by the brain's structure. These models consist of interconnected nodes arranged in layers. Other traditional ANN architectures used in PD detection include Self-Organizing Maps (SOM), which are used for clustering and visualization, and Learning Vector Quantization (LVQ) networks. ANNs have been widely applied for PD pattern recognition and classification of defect types. For example, ANNs have been used to classify discharges from different defects in ceramic insulators (Anjum et al., 2017) and to diagnose faults in motor stator windings (T. Chen et al., 2018). A Back-Propagation ANN (BP-ANN) has been used for PD identification in GIS based on acoustic emission signals (Hou et al., 2021), and a Learning Vector Quantization (LVQ) network has been used for localizing PD in noisy environments by analyzing winding sections (Mohammadirad et al., 2025). ANNs are selected for their ability to model complex, non-linear relationships between input features and output classes, as well as their capacity for learning directly from data (T. Chen et al., 2018; Hou et al., 2021). They have demonstrated high recognition rates for various PD faults and the capability to determine degradation stages (Mas'ud et al., 2016). In terms of performance, recognition rates exceeding 95% were reported for

each defect class using ANNs (Anjum et al., 2017). An accuracy of 90% was achieved in diagnosing stator faults (T. Chen et al., 2018), and 96.03% accuracy was obtained for insulator defect classification using discrete wavelet transform features (Haiba & Eliwa Gad, 2024). A study classifying PD types reported a 97% recognition rate (Zahed et al., 2017), and a BP-ANN model for GIS defects achieved 93.7% accuracy (Hou et al., 2021).

2.2.6 Fuzzy Logic Based Classifiers

Fuzzy logic systems operate on "degrees of truth" rather than binary true/false logic, making them well-suited for managing the uncertainty and imprecision inherent in real-world data. Common fuzzy logic-based classifiers include Fuzzy Decision Trees, Fuzzy C-Means (FCM) clustering, and Adaptive Neuro-Fuzzy Inference Systems (ANFIS). Fuzzy logic has been applied to diagnose PD in cast-resin transformers (Lee, 2020) and for rejecting interference (D. Wang et al., 2019). It is also used for identifying PD signals through clustering (B. Chen et al., 2017). An Adaptive Neuro-Fuzzy Inference System (ANFIS), which integrates neural networks with fuzzy logic, has been used for evaluating PD intensity by leveraging its ability to continuously optimize model parameters and reduce errors compared to traditional methods (D. Wang et al., 2019). The adaptability and robustness of ANFIS make it a strong candidate for PD analysis. Fuzzy logic methods are particularly valuable because they can be designed to provide interpretable, rule-based outputs, which aids in understanding the diagnostic decision-making process. Regarding performance, the Fuzzy Logic Clustering Decision Tree (FLCDT) was found to perform significantly better than traditional decision trees in terms of classification precision (Lee, 2020). An adaptive Fuzzy C-Means (FCM) method reported high accuracy in identifying PD signals (B. Chen et al., 2017). Furthermore, an improved ANFIS model showed a 2% reduction in model error compared to a traditional ANFIS when evaluating PD intensity (D. Wang et al., 2019).

2.2.7 Clustering Algorithms (K-Means, GMM, DBSCAN, etc.)

Clustering algorithms are unsupervised learning techniques that group objects such that those in the same cluster are more similar to each other than to those in different clusters. Common methods include K-Means, Gaussian Mixture Model (GMM), and density-based approaches like DBSCAN. Self-Organizing Maps (SOM) can also be viewed as a clustering technique. These algorithms have been applied to separate signals from multiple PD sources (Firuzi et al., 2017; Mishra et al., 2019) and perform data cleaning by separating signal from noise (Soh et al., 2022). They can also identify distinct PD patterns for localization (K. Zhou et al., 2021). Methods like K-Means, GMM, and Mean-shift have been combined to classify PD signals effectively (Carvalho et al., 2024). Self-Organizing Maps, when paired with statistical feature transformation, have been employed for PD classification (Bohari et al., 2021). A notable advancement is the development of PDAutoClust, a parameter-free clustering technique designed to reduce reliance on user-defined parameters, making the process more robust and accessible (Rahman et al., 2024). Clustering algorithms are valuable because they can discover underlying structures in PD data without requiring pre-labeled datasets. This is crucial for unsupervised monitoring where fault types may not be known in advance. GMM clustering helps discover distinct groups in a feature space for separating mixed PD signals (Firuzi et al., 2017), while density-based methods can effectively separate PD pulses from various types of noise and interference (K. Zhou et al., 2021). Performance-wise, a combination of K-Means, GMM, and Mean-shift clustering followed by an SVM classifier achieved 93% accuracy (Carvalho et al., 2024). The PDAutoClust method performed favorably against other density-based algorithms on PD datasets (Rahman et al., 2024). A SOM-based classifier demonstrated classification times under five seconds with nearzero error rates (Bohari et al., 2021), and an advanced density peak clustering algorithm reported an average localization error of just 0.21 m for two simultaneous PD sources (K. Zhou et al., 2021).

2.3 Feature Engineering in partial discharge Analysis

2.3.1 Time-Domain Features

Time-domain features are derived directly from the individual PD pulse waveforms to quantify their fundamental physical properties. These features characterize the shape, magnitude, and temporal occurrence of the pulse, which are intrinsically linked to the underlying discharge physics and provide crucial information for classification (Lu et al., 2020). The analysis of these characteristics enables the differentiation of various PD phenomena based on how the discharge manifests as a transient electrical signal (Boczar et al., 2017). The peak amplitude, signifying the maximum magnitude of the pulse, serves as a primary indicator of the discharge intensity and is directly related to the energy released during the event (S. Zhang et al., 2024). Different PD sources, such as corona, surface discharges, or internal voids, often generate pulses with distinct amplitude ranges. This makes amplitude a valuable feature for assessing a defect's severity and for distinguishing between source types, as the physical mechanism and geometry of the defect govern the amount of charge mobilized in the discharge (Boczar et al., 2017). The pulse rise time, defined as the interval for the signal to increase from 10% to 90% of its peak value, reflects the speed of the ionization process (Lu et al., 2020). PD pulses are characterized by very rapid rise times, often in the nanosecond or subnanosecond range (Darwish et al., 2019). This parameter is highly effective for discriminating between PD sources because phenomena such as streamer-like corona discharges exhibit different development speeds compared to Townsend-like discharges in internal voids (Xavier et al., 2021). Consequently, rise time is a critical feature, though it can be affected by signal degradation and dispersion during propagation from the source to the sensor (Lu et al., 2020). Pulse duration, frequently measured as the full width at half maximum (FWHM), quantifies the time for which the discharge event persists and is related to the recombination time of charge carriers following the initial ionization (Martínez-Tarifa et al., 2022). This temporal characteristic is another key descriptor of the pulse shape that aids in distinguishing between defect types. Specific

discharge mechanisms, influenced by factors like gas pressure and electrode geometry, produce characteristically longer or shorter pulses, making duration a useful parameter for classification (Boczar et al., 2017; Uwiringiyimana et al., 2022). Finally, the pulse count per cycle tallies the total number of PD events detected within one period of the power frequency, directly quantifying the repetition rate and activity level of the PD source (Gillis et al., 2023). A high pulse count can indicate a more active or severe insulation defect, and monitoring this metric is essential for tracking the evolution and stability of a fault over time (Ma et al., 2021). The number of discharges per cycle, along with their phase distribution, is fundamental to forming Phase-Resolved partial discharge (PRPD) patterns, which are widely used for identifying specific defect types (Boczar et al., 2017; Kasinathan et al., 2017).

2.3.2 Statistical Features

Statistical features provide a quantitative description of the shape and distribution of discharge activity within the Phase-Resolved partial discharge (PRPD) pattern. These descriptors are derived from the distribution of pulse magnitudes or counts across the phase angles of the AC voltage cycle. For this work, the higher-order statistical moments of skewness and kurtosis were computed, as they are particularly effective for capturing the nuanced characteristics of PRPD patterns that differentiate various discharge phenomena (Boczar et al., 2017). Skewness measures the degree of asymmetry in the distribution of PD pulses relative to the center of the phase axis. This feature is valuable because the physical mechanisms governing PD can differ significantly between the positive and negative half-cycles of the applied voltage, leading to characteristically asymmetric PRPD patterns (Boczar et al., 2017). For instance, corona from a sharp point electrode typically produces a much higher discharge activity during one polarity of the AC cycle, resulting in a highly skewed distribution (Yao et al., 2016). By quantifying this asymmetry, skewness provides a powerful metric for distinguishing between sources like corona, surface, and internal void discharges, each of which exhibits unique

symmetry properties in its PRPD profile (Bohari et al., 2021). Kurtosis quantifies the “peakedness” or “tailedness” of the distribution, indicating whether the PD pulses are concentrated within narrow phase windows or are more broadly dispersed across the cycle. A high kurtosis value signifies a distribution with sharp, narrow peaks and heavy tails, which is characteristic of discharge types that occur only at specific, repeatable moments near the voltage peaks (Boczar et al., 2017). Conversely, a lower kurtosis suggests a flatter, more uniform distribution of pulses. Since different insulation defects generate discharges with distinct phase concentrations, kurtosis serves as a key feature for identifying the nature of the PD source by characterizing the shape of its statistical distribution (Kunicki et al., 2018). Together, these statistical moments transform the qualitative visual information of a PRPD pattern into a compact and discriminative numerical feature vector suitable for automated classification (Bohari et al., 2021).

2.3.3 Frequency-Domain Features

Frequency-domain analysis is employed to investigate the spectral composition of partial discharge signals, which provides critical diagnostic information not apparent in the time domain. By applying a Fast Fourier Transform (FFT) to each captured waveform, the signal is converted from its temporal representation to its constituent frequencies (M. H. Li et al., 2018). The significance of this analysis lies in the fact that PD pulses are inherently fast, transient events with rapid rise times, meaning they are broadband phenomena containing significant energy across a wide spectrum that can extend into the High Frequency (HF), Very High Frequency (VHF), and Ultra-High Frequency (UHF) ranges (Darwish et al., 2019). The frequency signature of a measured PD signal is shaped by two key factors: the physical nature of the discharge source and the propagation path the electromagnetic wave travels to the sensor (Azirani et al., 2021). The equipment itself, such as a power transformer or a Gas-Insulated Switchgear (GIS) enclosure, acts as a complex waveguide, causing frequency-dependent attenuation and reflection that alters the signal’s spectral characteristics (Promwong & Tiengthong,

2020). Consequently, the energy distribution across different frequency bands becomes a unique fingerprint for a specific defect type within a given apparatus. Different types of insulation defects are known to manifest distinctively in the frequency domain. The UHF band (300 MHz–3 GHz) is particularly valuable for PD detection in transformers and GIS, as signals in this range can propagate efficiently within the metallic enclosures while being less susceptible to external corona and broadcast interference (Salah et al., 2022). For instance, the spectral content of UHF signals can be analyzed to diagnose and distinguish between specific fault types in GIS, such as those caused by free-moving particles or fixed protrusions on conductors (Ren et al., 2021). Selecting an optimal frequency band for analysis is therefore crucial for maximizing the signal-to-noise ratio and detection sensitivity (Azirani et al., 2021). For this study, the spectral power was computed within the distinct HF, VHF, and UHF bands to create a feature that captures this discriminative energy distribution, providing a robust basis for classification (Maneerot et al., 2019).

2.3.4 Time-Frequency Features

Partial discharge signals are fundamentally transient and non-stationary phenomena, meaning their frequency content evolves rapidly over their short duration. This characteristic makes time-frequency analysis methods, which can capture both temporal and spectral information simultaneously, particularly well-suited for their characterization (M. H. Li et al., 2018). Among these methods, the Discrete Wavelet Transform (DWT) is exceptionally effective and widely employed in PD analysis for its dual capability of signal denoising and feature extraction (Lu et al., 2020). The DWT decomposes a signal into various frequency sub-bands at different resolutions, allowing for the effective isolation of PD-related components from background noise and interference (Sahnoune et al., 2024). The efficacy of the DWT is highly dependent on the choice of the mother wavelet, as its shape should ideally correlate with the morphology of the PD pulses being analyzed to achieve optimal decomposition (Sahnoune et al., 2024).

For this study, the Daubechies 4 (db4) wavelet was selected. The Daubechies family of wavelets is frequently chosen for its properties of compact support and orthogonality, and the db4 wavelet, in particular, has been demonstrated in the literature to be effective for the denoising and analysis of PD signals, thereby representing an established and validated choice (Y. Wang et al., 2020). The signal was decomposed into four levels, which separates the signal into a series of detail coefficients representing successively narrower high-frequency bands, and a final approximation coefficient representing the low-frequency content. For feature extraction, the energy of the detail coefficients at each of the four decomposition levels was calculated. The resulting four-dimensional vector of wavelet energies serves as a powerful feature set. Since different types of PD sources (e.g., surface, corona, void) generate pulses with distinct spectral signatures, their energy is distributed differently across these frequency bands (Gao et al., 2023).

2.4 Dataset Diversity and Model Generalizability

The generalizability of machine learning (ML) models for partial discharge (PD) detection is fundamentally challenged by the immense diversity of PD data (Lu et al., 2020). Signal characteristics are highly sensitive to the defect type (Bin et al., 2020; Lee, 2020; Meitei et al., 2020; Zang et al., 2022), and operational environment. Consequently, a model trained on a specific dataset, particularly laboratory-generated data, often fails to generalize to different realworld scenarios (K. Sun et al., 2019; Raymond et al., 2022). To mitigate this, researchers employ strategies like domain adaptation (Tai et al., 2024; V. N. Tuyet-Doan et al., 2024; Niu & Zhu, 2025), and data augmentation to improve model robustness (Raymond et al., 2022; Rauscher et al., 2024). Despite these efforts, a critical gap persists: many studies still rely on limited, homogeneous datasets from controlled lab settings (W. Zhang et al., 2016). The scarcity of real-world data and the absence of standardized public benchmarks severely hinder the development of truly robust and widely applicable ML solutions (D. U. Kim et al., 2020; Lu et al., 2020).

2.5 Comparative Studies and Benchmarking in ML for partial discharge Detection

Comparative studies and benchmarking are essential for identifying the most effective algorithms and methodologies in ML-based PD detection. Several studies have undertaken such comparisons, revealing performance differences among models. For example, (Bin et al., 2020) found that a PSO-ELM outperformed SVM and k-NN for GIS defect classification. (Lee, 2020) showed their Fuzzy Logic Clustering Decision Tree surpassed standard decision trees for transformer diagnostics. Likewise, (Ma et al., 2021) demonstrated that SVM was superior to k-NN, Naive Bayes, and Decision Trees for classifying PD severity. However, existing comparative analyses often suffer from inconsistencies that make direct, field-wide comparisons challenging (Lu et al., 2020). A primary issue is the use of different, often private, datasets, which limits the generalizability of findings (J. Kim & Kim, 2021). Since PD signal characteristics vary significantly with the defect type, insulation, and equipment, results from one dataset are not directly applicable to another (Ranjan et al., 2024). A key limitation is that comparisons often focus on algorithmic performance without deeply exploring how feature engineering strategies interact with the choice of model (Govindarajan, Muthukumar, et al., 2021). The optimal ML algorithm for a given task might change depending on the features extracted from the raw signal. There is a notable lack of standardized benchmarking protocols that test across various feature-algorithm pairings and generalization scenarios (Lu et al., 2020). This need for standardization was highlighted in the development of a qualification procedure for PD analysers (Vera et al., 2023). Addressing these issues is essential for developing truly robust and reliable ML-based PD detection systems.

2.6 Research Gaps

Despite the promise of machine learning (ML) for partial discharge (PD) detection, a review of the literature reveals critical gaps that hinder the development of robust and

generalizable diagnostic tools. A primary gap is the limited exploration of featurealgorithm interactions; most studies evaluate ML models using a single, predefined set of features (Lu et al., 2020), neglecting to systematically investigate how different feature combinations impact the performance of various algorithms (Nimmo et al., 2017; Govindarajan, Muthukumar, et al., 2021). Furthermore, many models are trained and tested on limited, homogeneous datasets from a single equipment type or condition (Woon et al., 2016; J. Kim & Kim, 2021). This practice raises significant questions about their ability to generalize to unseen PD sources and diverse operational environments, a key barrier to practical deployment (Lu et al., 2020). This research addresses these gaps by systematically comparing traditional ML models across diverse PD datasets. We will evaluate how various feature combinations affect each algorithm’s performance to identify optimal pairings, a need highlighted by the lack of such analysis in current literature (Govindarajan, Muthukumar, et al., 2021). Furthermore, we will explicitly test model generalizability using a combined dataset reflecting real-world conditions from various equipment like transformers, GIS, and cables (Bin et al., 2020; Meitei et al., 2020), aiming to develop more robust and reliable diagnostic tools.

CHAPTER 3

METHODOLOGY

This chapter describes the methodological steps used to investigate partial discharge (PD) signal classification. It presents the complete research workflow, including dataset acquisition and organization, signal preprocessing, feature engineering and expansion, feature selection, model training, and performance evaluation. In this study, PD identification is formulated as a multi-class classification task, where each class corresponds to a PD signal category defined by expert annotations and galvanic reference measurements provided with the field dataset.

3.1 Overview of the Methodological Workflow

The study follows a structured processing pipeline beginning with dataset ingestion and organization, followed by preprocessing and signal conditioning. Feature engineering is then performed through base feature extraction and feature expansion. Subsequently, feature selection is conducted using two parallel selection tracks. Supervised machine learning models are trained using stratified cross-validation, and model performance is evaluated using standardized classification metrics. Each stage of the workflow is described in detail in the following sections, covering the dataset, preprocessing, feature engineering, feature selection, model training, and performance evaluation.

3.2 Datasets

This study utilizes a publicly available dataset obtained from long-term online monitoring of medium-voltage (MV) overhead power lines. The data were collected using a contactless antenna-based measurement technique designed to capture electromagnetic emissions associated with partial discharge activity in XLPE-covered conductors installed on MV overhead transmission lines (Klein et al., 2024). Antenna-based PD monitoring is a non-invasive radiometric approach that enables remote signal acquisition without direct electrical connection to high-voltage components (Cruz et al., 2023; Azam et al., 2024).

The dataset spans nearly three years of continuous measurements sampled at one-hour intervals from nine monitoring stations located in Czechia and Slovakia. Each data sample consists of a 20 ms time-domain waveform recorded at a sampling frequency of 1 MHz. Station identifiers and measurement identifiers are used to organize the data and to associate each waveform with its corresponding label provided in the annotation file.

3.3 Data Preprocessing and Signal Conditioning

Preprocessing converts raw waveforms into a standardized form suitable for feature computation and supervised learning. The preprocessing pipeline consists of dataset discovery, bandpass filtering, normalization, and label assignment, and is applied uniformly to all signals prior to feature extraction.

3.3.1 Signal Acquisition and Discovery

All measurement stations are automatically discovered from the dataset directory structure. Each station directory follows the naming pattern `station_<ID>` and contains raw waveform files stored in NumPy array format (`.npy`). The preprocessing pipeline iterates through all station directories and loads each waveform sequentially in order to support large-scale processing without exceeding memory constraints.

3.3.2 Bandpass Filtering

Each waveform is filtered using a fourth-order Butterworth bandpass filter to retain the frequency range of interest while suppressing out-of-band components. Filtering is implemented using a forward-backward zero-phase approach through the `filtfilt` routine (K. Sun et al., 2019). The sampling frequency is set to 1 MHz, with a lower cutoff frequency of 1 kHz and an upper cutoff frequency of 450 kHz, corresponding to 45% of the Nyquist frequency.

3.3.3 Normalization

After filtering, each signal is standardized using z-score normalization according to

$$x_{\text{normalized}} = \frac{x - \mu}{\sigma}, \quad (3.1)$$

where μ and σ denote the sample mean and standard deviation, respectively. For constant-valued signals where $\sigma = 0$, the standard deviation is set to 1.0 to avoid division by zero. All normalized waveforms are stored using 32-bit floating-point representation to reduce memory usage while maintaining numerical precision.

3.3.4 Metadata Integration and Label Assignment

For each processed waveform, metadata including station identifier, measurement identifier, signal length, and file path are recorded. Class labels are assigned by matching each waveform to the expert annotation table `inferred_annotation.csv` using the station and measurement identifiers as keys. If no matching annotation is found, the waveform is assigned the label `unknown`. Metadata and labels are stored in Parquet format, while the processed waveforms are saved as individual `.npy` files for subsequent feature extraction.

3.4 Feature Engineering

Feature engineering transforms each preprocessed waveform into a numerical representation suitable for supervised classification. The feature engineering pipeline consists of two stages: base feature extraction and feature expansion. All features are computed from the complete 20 ms waveform without applying additional segmentation or windowing.

3.4.1 Base Feature Extraction from Cleaned Data

Each preprocessed waveform is loaded from its stored `.npy` file and processed sequentially. Given a one-dimensional normalized and bandpass-filtered signal sampled at 1 MHz, a feature vector is computed for each waveform. The resulting feature vectors are aggregated into a feature matrix, where each row corresponds to a waveform and each column corresponds to a feature. Station identifiers, measurement identifiers, and class labels are preserved alongside the feature matrix to maintain traceability.

3.4.2 Base Feature Types

A total of 23 base features are computed across multiple domains, including time-domain statistics, frequency-domain descriptors, time–frequency descriptors, entropy measures, a fractal dimension estimate, and phase-resolved partial discharge (PRPD) summary features.

3.4.2.1 Time-Domain Features

Time-domain features are computed directly from the waveform samples $\{x_i\}_{i=1}^N$. The root mean square value is calculated as

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}.$$

Skewness and kurtosis are computed using unbiased estimators implemented through the `scipy.stats` library, with NaN values replaced by zero. The crest factor is calculated as the ratio of the maximum absolute amplitude to the RMS value, with the denominator set to 1.0 when the RMS value is zero. The signal variance is computed as

$$\text{var} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2.$$

3.4.2.2 Frequency-Domain Features

Frequency-domain features are derived from the magnitude spectrum obtained using the real-valued fast Fourier transform (`np.fft.rfft`). The mean FFT magnitude is computed as the average magnitude across all frequency bins, while the maximum FFT magnitude corresponds to the largest spectral component.

3.4.2.3 Time–Frequency Features

Time–frequency features are computed using the short-time Fourier transform implemented via `scipy.signal.stft`. The segment length is adaptively selected as the minimum of 256 and the maximum of 8 and the signal length. The mean and maximum magnitudes across the resulting time–frequency representation are retained as features.

3.4.2.4 Entropy Features

Entropy-based features are computed to summarize signal distributions. Shannon entropy is calculated from a discretized amplitude histogram with 64 bins and density normalization, including a small regularization term to avoid numerical instability. Spectral entropy is computed using the normalized FFT magnitude spectrum and the same entropy formulation.

3.4.2.5 Fractal Dimension

The Higuchi fractal dimension is computed using a maximum scale parameter of $k_{\max} = 5$. The fractal dimension is estimated from the slope of the linear regression between $\log(1/k)$ and $\log(\langle L(k) \rangle)$, where $\langle L(k) \rangle$ denotes the average curve length computed at scale k .

3.4.2.6 PRPD Features

Phase-resolved partial discharge features are computed by mapping waveform samples to phase angles assuming a mains frequency of 50 Hz according to

$$\phi = \left(\frac{t}{f_s} \right) \times 50 \times 2\pi \bmod 2\pi.$$

The phase cycle is divided into four equal quadrants. For each quadrant, the mean absolute amplitude and sample density are computed. Additional PRPD summary features include the polarity ratio between positive and negative samples, the peak absolute amplitude, and the normalized signal energy.

3.4.3 Feature Expansion

Following base feature extraction, additional derived features are generated through mathematical combinations of feature pairs. Prior to expansion, metadata columns are removed so that only numerical features are processed. Feature expansion includes pairwise arithmetic combinations, second-degree interaction terms generated using scikit-learn polynomial feature transformations, and additional mathematical combinations computed using the `feature-engine` library. Duplicate features are removed after expansion, and the final feature matrix is formed by concatenating the base and derived features.

3.5 Feature Selection

Two parallel feature selection tracks are implemented to produce reduced feature sets for model training. Each track outputs the selected feature names and associated

metadata, which are stored in JSON format for reproducibility.

3.5.1 Track 4B: Featurewiz

The first feature selection track applies the Featurewiz library. Correlation-based pruning is first performed using the SULOV procedure with a correlation threshold of 0.7. XGBoost-based feature importance ranking is then applied to the remaining features to obtain a reduced feature subset. If Featurewiz is unavailable or fails at runtime, a variance-based filter is applied as a fallback.

3.5.2 Track 4C: MLJAR AutoML

The second feature selection track uses MLJAR-supervised AutoML with feature selection enabled. The AutoML process operates in `Explain` mode with a total time limit of 600 seconds and a per-model time limit of 120 seconds. Feature selection is integrated into the internal cross-validation procedure, and the selected features are extracted from the best-performing AutoML model.

3.6 Model Training and Cross-Validation

Machine learning models are trained separately for each feature selection track using stratified cross-validation. All training outputs, including fold-level predictions and probability estimates, are retained for evaluation.

3.6.1 Data Preparation

For each track, the selected feature list is used to subset the feature matrix. Meta-data columns are separated from numerical features, and the feature matrix is converted to 32-bit floating-point format. Target labels are encoded using scikit-learn `LabelEncoder`.

3.6.2 Cross-Validation Strategy

Stratified k-fold cross-validation is employed to preserve class proportions across folds. The number of folds is determined by the minimum class count, with a maximum of 10 folds and a minimum of 5 folds when feasible. When the minimum class count is less than two, a single stratified shuffle split is used with a 70/30 training-validation split. All splitters use a fixed random seed for reproducibility.

3.6.3 Model Specifications

Five supervised classifiers are trained: a support vector machine with an RBF kernel, a random forest classifier, a decision tree classifier, a k-nearest neighbors classifier, and an artificial neural network implemented using a multilayer perceptron architecture. All models are configured using fixed hyperparameters and a consistent random seed.

3.6.4 Training Procedure

For each classifier and cross-validation fold, a new model instance is trained using the training partition and evaluated on the validation partition. Predictions and class probabilities are recorded for each fold. Performance metrics are computed at the fold level and aggregated by calculating the mean and standard deviation across folds. A final model for each classifier is trained using the full dataset and stored together with its associated metadata.

3.7 Performance Evaluation and Reporting

Model performance is evaluated using cross-validation outputs for each feature selection track. Summary metrics and visual diagnostics are generated to support comparative analysis.

3.7.1 Evaluation Metrics

Accuracy, precision, recall, F1-score, and ROC-AUC are computed using macro-averaging. For multi-class evaluation, the one-vs-rest strategy is used for ROC-AUC computation. For each metric, the mean and standard deviation across cross-validation folds are reported.

3.7.2 Visualization and Reporting

Evaluation outputs include confusion matrices, multi-class ROC curves, and comparative charts of F1-scores across models. All metrics, predictions, and visualizations are stored in track-specific directories and saved in multiple formats, including Markdown, CSV, JSON, and Parquet, to support reproducibility and further analysis.

CHAPTER 4

RESULTS

4.1 Experimental Results

This chapter reports the classification results obtained from high frequency partial discharge (PD) signals acquired from medium voltage overhead lines. Two optimization regimes are compared in terms of their selected feature sets and the resulting classification performance a Featurewiz-based selection track and an MLJAR track. Model performance is reported using Accuracy, Precision, Recall, F1-score, and ROC AUC, with results summarized in the tables and figures that follow.

4.2 Comparison of Feature Selection Outcomes

The first set of results compares the features retained by the two optimization regimes. Table 4.1 lists the 30 features that were jointly selected, providing a direct view of the overlap between the Featurewiz and MLJAR selections. The shared set spans time-domain statistics (for example, skewness, kurtosis, and crest_factor), frequency-domain descriptors (for example, fft_mean, fft_max, stft_mean, stft_max, and spectral_entropy), PRPD pattern measures (for example, multiple quadrant-based differences and interactions), and complexity measures (for example, fractal_dimension and entropy combinations). The prevalence of both original descriptors and interaction terms among the jointly selected features indicates that both regimes consistently retained information reflecting amplitude characteristics, spectral content, phase-resolved distribution structure, and signal complexity.

Table 4.1 Comparison of Jointly Selected Feature Families Across Featurewiz and MLJAR Feature Selection Methods

Category	Features	Feature Type	Featurewiz	MLJAR	Selection Rationale
Time Domain	Skewness, Kurtosis, Crest Factor	Original	Yes	Yes	Capture waveform asymmetry, impulsivity, and peak-to-RMS characteristics associated with discharge intensity
Time-Domain Statistical Interaction	RMS-Skewness and RMS-Kurtosis Interaction Ratios	Interaction	Yes	Yes	Relate signal amplitude to higher-order statistical moments, enhancing sensitivity to transient PD behavior
Frequency Domain	FFT Mean and FFT Maximum Spectral Magnitude	Original	Yes	Yes	Summarize global spectral energy distribution and dominant frequency components
Time-Frequency Domain	STFT Mean and STFT Maximum Energy	Original	Yes	Yes	Capture localized time-frequency energy concentrations associated with non-stationary PD activity
Entropy Measures	Spectral Entropy	Original	Yes	Yes	Quantifies spectral disorder and complexity under varying signal conditions
Time-Frequency Interaction	Global-Local Spectral Ratios (FFT/STFT)	Interaction	Yes	Yes	Relate global frequency content to localized transient behavior under interference
PRPD Pattern (Composite)	Quadrant-Wise PRPD Mean Differences and Absolute Differences	Composite Interaction	Yes	Yes	Encode phase-dependent asymmetry and discharge pattern shifts across the AC cycle using multi-combination PRPD statistics
PRPD Pattern (Composite)	PRPD Density, Energy, and Polarity Interaction Features	Composite Interaction	Yes	Yes	Capture structured phase-resolved discharge distributions using composite m_c feature interactions
Complexity Measures	Fractal Dimension	Original	Yes	Yes	Measures waveform self-similarity and geometric complexity
Entropy Interaction (Cross-Domain)	Combined Time-Frequency Entropy Measures	Interaction	Yes	Yes	Aggregate entropy information across domains to summarize non-stationary signal behavior

Overall, the overlap in Table 4.1 shows that both optimization regimes converged on a common set of informative descriptors across multiple feature categories. In addition to shared original features, the jointly selected interaction terms largely reflect ratios, differences, and absolute differences that relate global and localized spectral summaries and quantify contrasts between PRPD quadrants.

4.2.1 Top-Ranked Features Under Each Selection Track

Figure 4.1 presents the top ten features ranked by ANOVA F-score under the Featurewiz–XGBoost track. The ranking is dominated by `spectral_entropy`, followed by frequency-domain interaction and entropy-combination terms, including `fft_mean_div_stft_mean` and `shannon_entropy_plus_spectral_entropy`. Classical time-domain statistics such as kurtosis appear within the top ten but are ranked below entropy- and FFT/STFT-derived descriptors.

Figure 4.2 shows the corresponding top ten ranking produced by the MLJAR internal feature selection track. In contrast to the Featurewiz track, MLJAR prioritizes composite and interaction variables dominated by combinations of `spectral_entropy` with `fractal_dimension`, RMS-related terms, and PRPD-related ratios. Notably, `spectral_entropy` and its derived combinations appear consistently among the highest-ranked features across both tracks, indicating stable retention of entropy-based descriptors despite differences in the specific formulations emphasized by each regime.

Overall, the top-ranked features from both tracks show agreement at the level of feature families (entropy/complexity and frequency-domain summaries), while differing in the relative emphasis placed on standalone descriptors versus composite interaction terms.

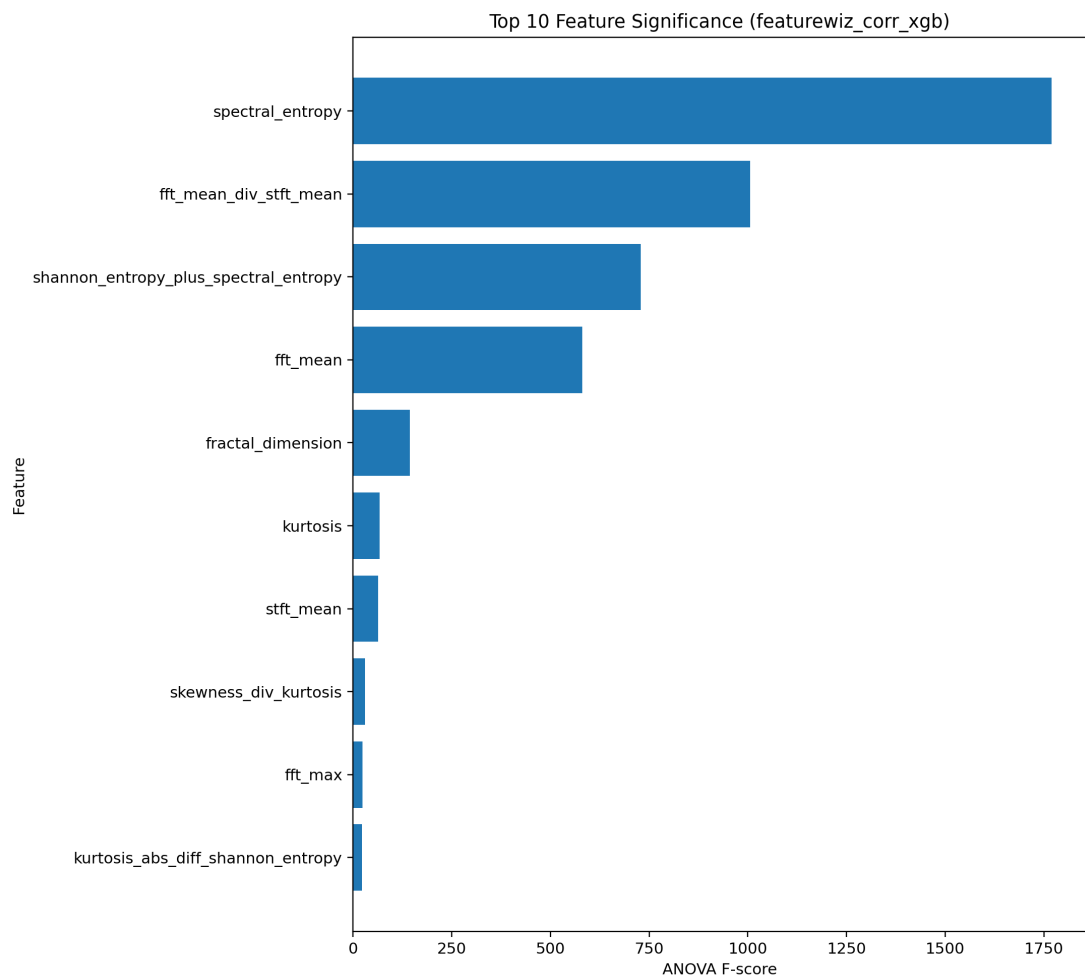


Figure 4.1 Top 10 Feature Significance (ANOVA F-Score) Under the Featurewiz–XGBoost Track

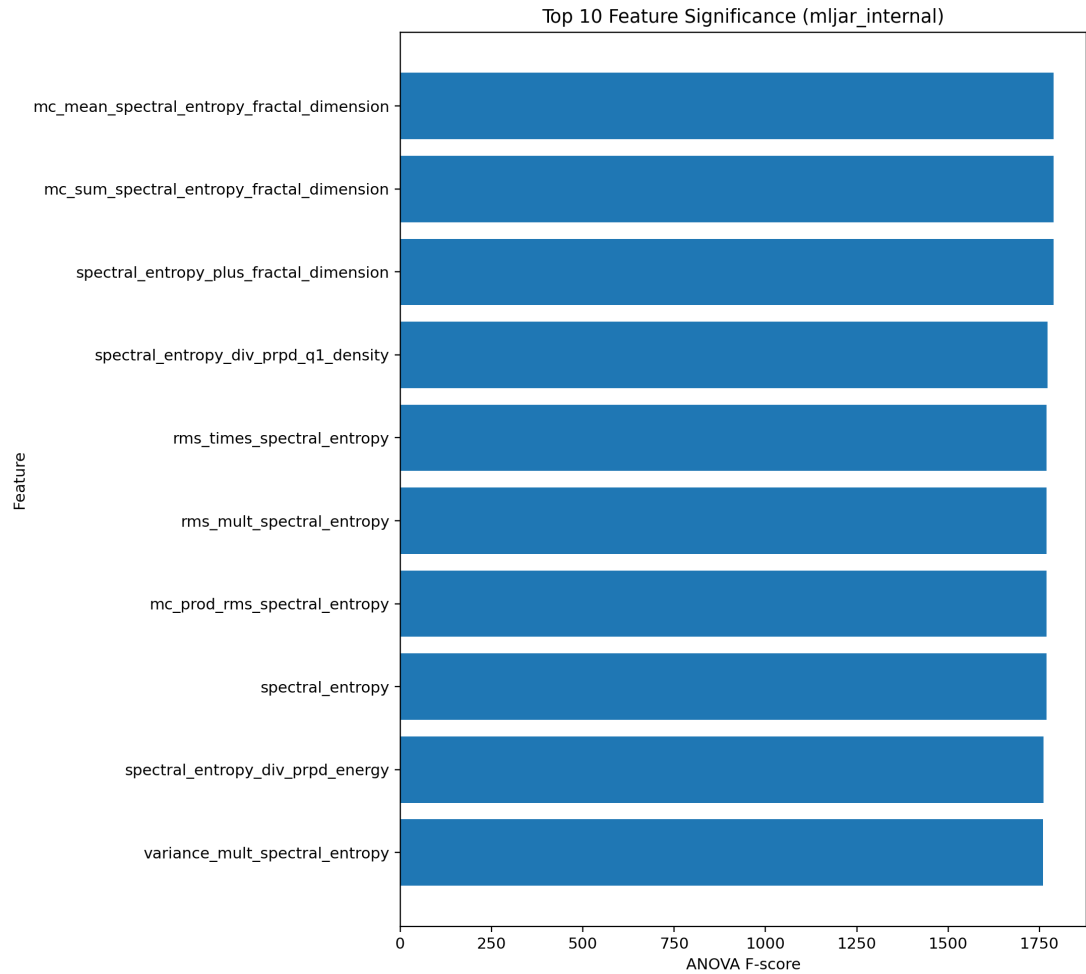


Figure 4.2 Top 10 Feature Significance (ANOVA F-Score) Under the MLJAR Internal Feature Selection Track

4.2.2 Comparative Analysis of Feature Selection Tracks

The feature selection outcomes are summarized as two tracks. Track 4B corresponds to the Featurewiz selection, resulting in 31 retained features, whereas Track 4C corresponds to the MLJAR feature set, comprising 2070 features. Table 4.2 reports the best F1 score achieved under each track, enabling a direct comparison between feature count and performance.

Table 4.2 Comparative Analysis of Feature Selection Tracks: Feature Count and Best Model F1-Score

Track ID	Selection Method	Feature Count	Best Model F1-Score (%)
Track 4C	MLJAR	2070	73.11 (Random Forest)
Track 4B	Featurewiz	31	71.84 (Random Forest)

As shown in Table 4.2, Track 4C achieved the highest reported F1 score (73.11%) with 2070 variables, while Track 4B achieved an F1 score of 71.84% with 31 variables. The absolute difference in F1 score between the two tracks is 1.27%.

4.3 Comparative Machine Learning Model Performance Evaluation

The classification performance of the models is summarized under both optimization regimes using the reported metrics.

4.3.1 Results with Featurewiz Selection

Table 4.3 summarizes the performance metrics for the classifiers when using the Featurewiz selected feature set (Track 4B). The results are reported as the average of a 10-fold stratified cross-validation process.

Table 4.3 Comprehensive Performance Evaluation Using Featurewiz Feature Selection (Track 4B)

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
RF	91.91 ± 0.62	83.55 ± 3.18	67.09 ± 2.25	71.84 ± 2.71	90.24 ± 0.75
SVM	91.64 ± 0.44	84.50 ± 2.39	64.36 ± 2.08	69.13 ± 2.54	88.18 ± 1.13
kNN	90.71 ± 0.47	77.89 ± 2.91	63.21 ± 1.40	67.05 ± 1.71	79.43 ± 2.30
ANN	89.42 ± 0.58	71.61 ± 1.76	69.21 ± 2.87	70.23 ± 2.31	86.33 ± 1.65
DT	87.21 ± 0.97	66.63 ± 2.27	67.49 ± 2.48	67.01 ± 2.30	67.49 ± 2.48

Note: RF = Random Forest, SVM = Support Vector Machine, kNN = k-Nearest Neighbors, ANN = Artificial Neural Network, DT = Decision Tree.

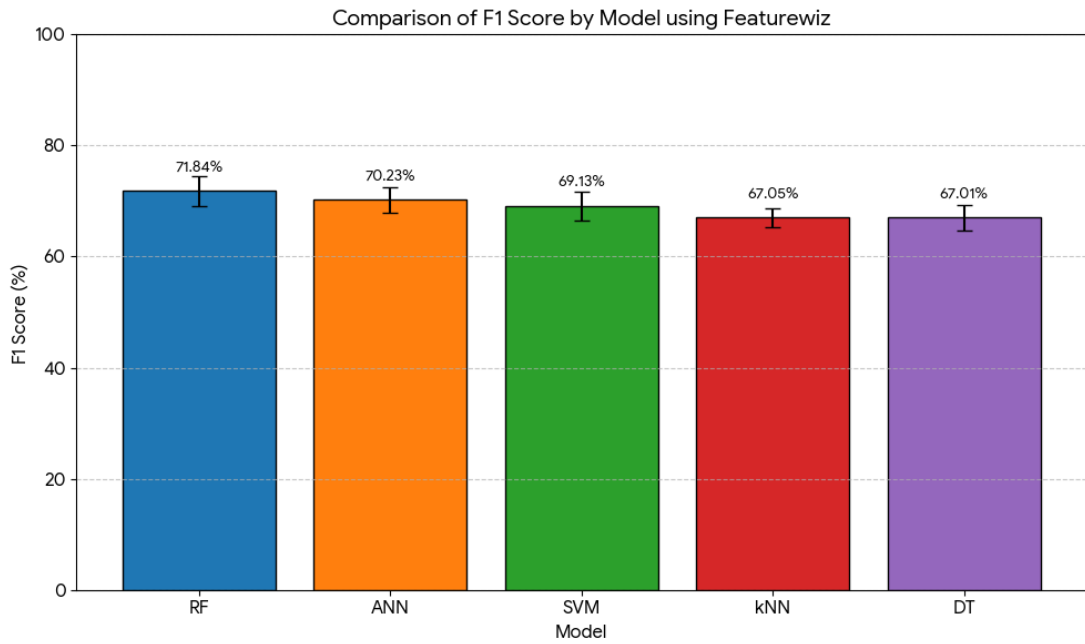


Figure 4.3 Bar Chart Visualization of Model Performance Metrics Using Featurewiz Feature Selection

As shown in Table 4.3, Random Forest achieved the highest F1 score (71.84%) within Track 4B, with an accuracy of 91.91% and a ROC AUC of 90.24%. SVM attained a similar accuracy (91.64%) and an F1 score of 69.13%, while ANN recorded the highest recall (69.21%) with an F1 score of 70.23%. Figure 4.3 provides a visual comparison consistent with the tabulated metrics.

Across the Featurewiz setting, accuracy values are concentrated around 90–92% for RF, SVM, and k-NN, whereas Decision Tree and ANN yield lower accuracies of 87.21% and 89.42%, respectively. Despite similar accuracies among RF, SVM, and k-NN, the F1 scores differentiate these models more clearly, reflecting differences in the reported precision and recall values.

Decision Tree produced the lowest ROC AUC (67.49%), while Random Forest recorded the highest F1 score (71.84%) alongside a high precision value (83.55%). These results indicate that within Track 4B, the strongest balance of precision and recall is achieved by Random Forest.

4.3.2 Results with MLJAR

Table 4.4 presents the results obtained using the MLJAR track (Track 4C). The results enable a direct comparison with the Featurewiz selection results in Table 4.3.

Table 4.4 Comprehensive Performance Evaluation Using MLJAR Feature Selection (Track 4C)

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
RF	91.83 $\pm$ 0.48	81.50 $\pm$ 1.93	68.94 $\pm$ 2.15	73.11 $\pm$ 2.23	90.63 $\pm$ 0.47
ANN	89.80 $\pm$ 1.11	73.41 $\pm$ 2.74	72.32 $\pm$ 1.49	72.65 $\pm$ 1.23	87.45 $\pm$ 1.50
kNN	90.97 $\pm$ 0.62	77.53 $\pm$ 2.82	67.46 $\pm$ 1.99	70.91 $\pm$ 2.12	82.07 $\pm$ 1.92
SVM	91.55 $\pm$ 0.40	84.49 $\pm$ 2.49	63.87 $\pm$ 2.10	68.54 $\pm$ 2.53	89.07 $\pm$ 0.87
DT	87.76 $\pm$ 0.67	68.09 $\pm$ 1.53	69.24 $\pm$ 1.51	68.62 $\pm$ 1.45	69.24 $\pm$ 1.51

Note: RF = Random Forest, ANN = Artificial Neural Network, kNN = k-Nearest Neighbors, SVM = Support Vector Machine, DT = Decision Tree.

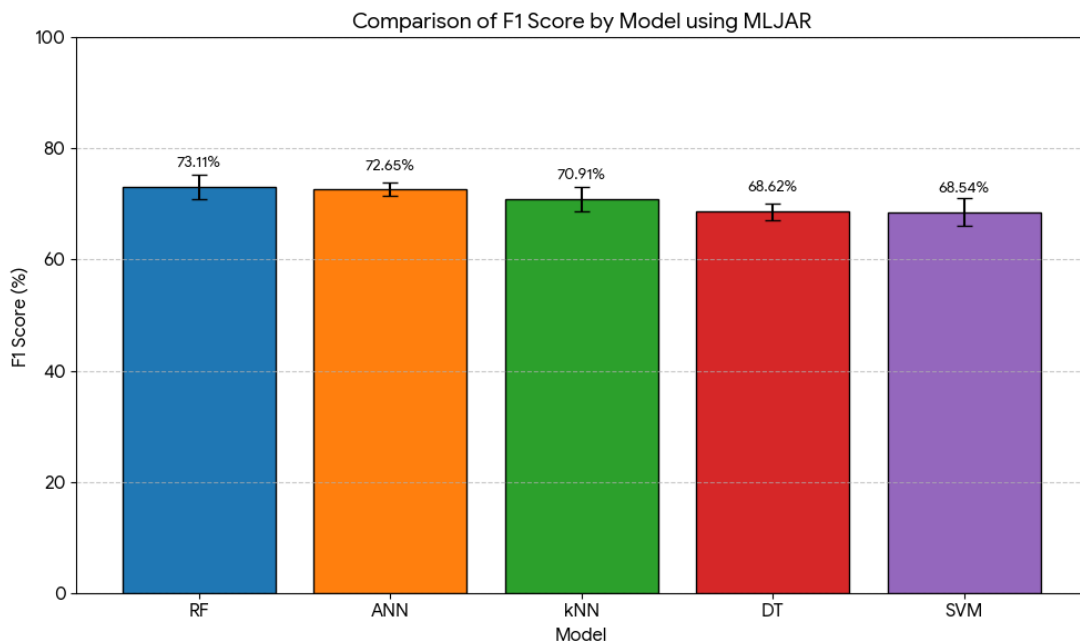


Figure 4.4 Bar Chart Visualization of Model Performance Metrics Using MLJAR Feature Selection

Table 4.4 presents the classification performance obtained using the MLJAR feature selection framework (Track 4C) for all evaluated machine learning models. Performance was assessed using 10-fold stratified cross-validation and reported in terms of accuracy, precision, recall, F1-score, and ROC-AUC, including their corresponding standard

deviations. These results enable direct comparison with the Featurewiz-based results reported in Table 4.3.

Among the evaluated models, Random Forest (RF) achieved the highest F1-score of $73.11 \pm 2.23\%$ and the highest ROC-AUC of $90.63 \pm 0.47\%$. The Artificial Neural Network (ANN) obtained an F1-score of $72.65 \pm 1.23\%$ and recorded the highest recall at $72.32 \pm 1.49\%$. The k-Nearest Neighbors (kNN) classifier achieved an F1-score of $70.91 \pm 2.12\%$, with accuracy and ROC-AUC values of $90.97 \pm 0.62\%$ and $82.07 \pm 1.92\%$, respectively.

Support Vector Machine (SVM) produced the highest precision among all models at $84.49 \pm 2.49\%$ and achieved an accuracy of $91.55 \pm 0.40\%$, while its F1-score was $68.54 \pm 2.53\%$. Decision Tree (DT) recorded the lowest overall performance, with an F1-score of $68.62 \pm 1.45\%$, accuracy of $87.76 \pm 0.67\%$, and ROC-AUC of $69.24 \pm 1.51\%$.

Figure 4.4 illustrates the comparative F1-score performance of all classifiers under the MLJAR feature selection framework and reflects the ranking observed in Table 4.4.

4.3.3 Performance Comparison of ML Models Under Different Feature Selection Techniques

Figure 4.5 visualizes the performance differences observed between the Featurewiz (Track 4B) and MLJAR (Track 4C) strategies.

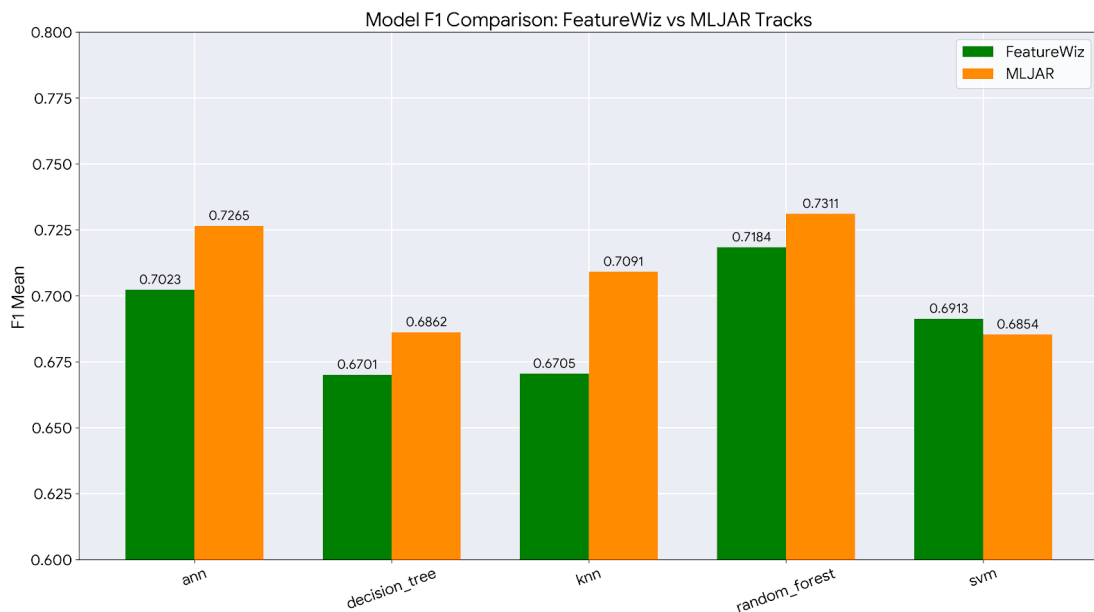


Figure 4.5 Comparative Performance Visualization of Models Under Featurewiz (Track 4B) and MLJAR (Track 4C) Feature Selection

Figure 4.5 presents a comparative visualization of the F1-score performance of machine learning models evaluated using Featurewiz (Track 4B) and MLJAR (Track 4C) feature selection strategies. The figure summarizes the mean F1-scores obtained for each classifier under both tracks.

Across all evaluated models, the MLJAR track yields higher F1-scores than the Featurewiz track for Artificial Neural Network (ANN), Decision Tree (DT), k-Nearest Neighbors (kNN), and Random Forest (RF). The Support Vector Machine (SVM) model shows comparable F1-score values across both feature selection approaches. The highest F1-score under the Featurewiz track is 71.84%, while the highest F1-score under the MLJAR track is 73.11%.

The comparative trends illustrated in Figure 4.5 directly reflect the mean F1-score values reported in Tables 4.3 and 4.4. The figure serves as a visual summary of the tabulated results, highlighting the relative performance differences between models under the Featurewiz and MLJAR feature selection frameworks.

4.4 Qualitative and Quantitative Comparison with Other Studies

Table 4.5 presents a focused qualitative and quantitative comparison between the present study and selected recent works on partial discharge (PD) classification that employ traditional machine learning techniques. The comparison is intentionally limited to studies published within the last five years to ensure contemporary relevance. Deep learning-based approaches are excluded to maintain methodological consistency and to enable fair benchmarking against classical machine learning classifiers.

Table 4.5 Benchmark Comparison of Partial Discharge Classification Studies with Explicitly Reported Metrics

Study	Machine Learning Models	Performance Metrics	Feature Selection / Utilization
(Almehdhar & Prochazka, 2025)	Deep Learning	F1-Score: 96.70%	Time-frequency analysis techniques for multiple PD source classification
(Ishaq et al., 2025)	Deep Learning, CNN	Accuracy: 100%	Signal-processing + deep-learning framework for PD classification
(Kalaivani et al., 2025)	Neural Network, CNN	Accuracy: 99%	Adversarial de-noising with enhanced feature extraction from PRPD patterns
(S. Li et al., 2025)	Deep Learning	Accuracy: 98%	CAE-ACGAN and ResNet-based PD pattern recognition under class imbalance
Our Methodology	MLJAR Random Forest	Accuracy: 91.83% Precision: 81.50% Recall: 68.94% F1-Score: 73.11%	Multi-domain features: Time, Frequency, Statistical, PRPD

As summarized in Table 4.5, recent partial discharge (PD) classification studies predominantly employ deep learning-based architectures, including convolutional neural networks and adversarial learning frameworks, to achieve high classification performance. The reported results indicate strong performance metrics, with accuracy values approaching or reaching near-perfect levels and high F1-scores, as observed in studies such as (Almehdhar & Prochazka, 2025; Ishaq et al., 2025; Kalaivani et al., 2025; S. Li et al., 2025). These outcomes are largely attributed to the capability of deep learning models to automatically extract hierarchical and discriminative representations from

complex PD signal patterns.

Despite their strong reported performance, the deep learning approaches summarized in the table rely on computationally intensive architectures, extensive data pre-processing, and large labeled datasets. Several studies further incorporate advanced techniques such as adversarial denoising, generative models, or deep feature augmentation to address challenges such as noise contamination and class imbalance. While these strategies enhance classification accuracy, they also increase model complexity and reduce transparency, which may pose challenges for practical deployment and physical interpretability in industrial PD monitoring applications.

In contrast, the proposed methodology adopts a classical machine learning framework based on a Random Forest classifier combined with automated feature selection across multiple PD domains. Although the achieved accuracy is lower than that reported by deep learning-based methods, the present work provides a more comprehensive and balanced evaluation through the simultaneous reporting of accuracy, precision, recall, and F1-score. Furthermore, the integration of time-domain, frequency-domain, statistical, and phase-resolved partial discharge (PRPD) features enables meaningful physical interpretation of the classification results. This positions the proposed approach as a transparent and computationally efficient alternative for PD diagnosis, particularly in scenarios where data availability, interpretability, and deployment constraints are critical considerations.

CHAPTER 5

DISCUSSION

5.1 Overview of Key Findings

This chapter provides a critical interpretation of the experimental results presented in Chapter 4, examining the performance characteristics of traditional machine learning algorithms for partial discharge classification and the influence of feature selection strategies on diagnostic accuracy (Aldosari et al., 2023; Govindarajan et al., 2023). The discussion addresses three fundamental aspects of the research: the comparative evaluation of machine learning algorithms under standardized experimental conditions, the identification and interpretation of optimal feature subsets through independent selection regimes, and the alignment of observed performance patterns with established literature in partial discharge diagnostics (Srivastava et al., 2023; Zemouri & Levesque, 2023). The analysis focuses on understanding the underlying mechanisms driving classification performance rather than merely reporting numerical outcomes, with particular attention to the trade-offs between different performance metrics and the physical interpretability of selected feature representations (Seitz et al., 2024; C. Zhang et al., 2024). This interpretive framework enables evidence-based conclusions regarding algorithm suitability for field deployment scenarios and provides insights into the stability and diagnostic relevance of multi-domain feature engineering approaches in partial discharge monitoring applications (Florkowski, 2024; Long et al., 2024).

5.2 Discussion Based on Research Objectives

5.2.1 Comparative Evaluation of Traditional Machine Learning Algorithms

The experimental results establish a clear performance hierarchy among the evaluated machine learning classifiers when applied to partial discharge classification under both feature selection regimes (Aldosari et al., 2023; Govindarajan et al., 2023). Under Track 4C (MLJAR feature selection), Random Forest achieved the highest F1 score of 73.11% alongside the highest ROC-AUC of 90.63%, with an accuracy of 91.83% and precision of 81.50%. Under Track 4B (Featurewiz selection), Random Forest similarly achieved the highest F1 score of 71.84% with a ROC-AUC of 90.24%, accuracy of 91.91%, and precision of 83.55%. Artificial Neural Networks demonstrated the highest recall performance across both tracks, achieving 72.32% on Track 4C and 69.21% on Track 4B, with corresponding F1 scores of 72.65% and 70.23% respectively (Aldosari et al., 2023; Srivastava et al., 2023). Support Vector Machines achieved the highest precision values, reaching 84.50% on Track 4B and 84.49% on Track 4C, with accuracy values of 91.64% and 91.55% respectively, and F1 scores of 69.13% and 68.54% (Govindarajan, Ardila-Rey, et al., 2021; Aldosari et al., 2023).

The superior performance of Random Forest classifiers across both feature selection tracks indicates that ensemble methods provide optimal balance between precision and recall for partial discharge classification tasks (Aldosari et al., 2023; Govindarajan et al., 2023). This outcome demonstrates that the complex, non-linear relationships inherent in partial discharge signals are more effectively captured through ensemble averaging of multiple decision trees rather than through single-tree models or linear margin-based classifiers (Govindarajan, Ardila-Rey, et al., 2021; Srivastava et al., 2023). The ability of Random Forest to maintain robust performance across substantially different feature set sizes (31 features versus 2070 features) suggests that its ensemble mechanism provides inherent regularization that mitigates overfitting risks associated with high-dimensional feature spaces (Aldosari et al., 2023; Zemouri & Levesque, 2023). This finding validates

the research objective of identifying the most suitable machine learning algorithm for partial discharge classification, as Random Forest consistently outperformed alternative approaches while maintaining interpretability through feature importance rankings (Govindarajan et al., 2023; Seitz et al., 2024). The observed performance advantage of Artificial Neural Networks in recall metrics, despite lower overall F1 scores compared to Random Forest, indicates that neural networks can effectively exploit higher-order feature interactions and complex decision boundaries that enable improved detection sensitivity for minority class instances (Aldosari et al., 2023; Srivastava et al., 2023). However, the superior balanced performance of Random Forest, as evidenced by higher F1 scores, suggests that ensemble methods provide a more optimal trade-off between detection sensitivity and prediction confidence (Govindarajan et al., 2023; Zemouri & Levesque, 2023).

The observed dominance of Random Forest in the present study aligns with its widespread adoption in partial discharge pattern recognition research utilizing engineered feature representations (Aldosari et al., 2023; Govindarajan et al., 2023). Previous studies have similarly reported strong performance of Random Forest classifiers in PD diagnostics, with applications ranging from switchgear fault diagnosis to cable insulation assessment (Govindarajan, Ardila-Rey, et al., 2021; Seitz et al., 2024). The consistency between the present results and prior literature suggests that ensemble tree methods represent a robust algorithmic choice for partial discharge classification tasks, particularly when working with multi-domain feature representations extracted from field monitoring data (Aldosari et al., 2023; Long et al., 2024). The strong performance of Support Vector Machines observed in this study, achieving precision values exceeding 84% across both tracks, is consistent with their frequent application in PD source classification and defect type recognition (Govindarajan, Ardila-Rey, et al., 2021; Aldosari et al., 2023). Previous research has demonstrated SVM effectiveness in high-dimensional feature spaces, which aligns with the present findings where SVM maintained competitive performance despite the substantial dimensionality differences between Track 4B (31 features) and Track 4C (2070 features) (Govindarajan et al., 2023; Zemouri & Levesque, 2023). However, the present results indicate that while SVM provides

excellent precision, Random Forest achieves superior balanced performance through higher F1 scores, suggesting that ensemble methods may be preferable when both detection sensitivity and prediction confidence are operationally important (Govindarajan et al., 2023; Adhikari et al., 2024). The observed performance of Artificial Neural Networks, achieving the highest recall values but lower overall F1 scores compared to Random Forest, reflects a pattern that has been reported in previous PD diagnostics research (Aldosari et al., 2023; Srivastava et al., 2023). Prior studies have noted that neural networks can effectively capture complex feature interactions but may require careful regularization and hyperparameter tuning to achieve optimal balanced performance (Aldosari et al., 2023; Zemouri & Levesque, 2023). The present results extend this understanding by demonstrating that while neural networks excel in recall-oriented applications, ensemble methods provide superior overall performance when multiple metrics must be simultaneously optimized (Govindarajan et al., 2023; Srivastava et al., 2023).

5.2.2 Identification of Optimal Feature Sets via Feature Selection

The feature importance rankings obtained from the Featurewiz–XGBoost and MLJAR internal selection frameworks reveal an apparent divergence in the prioritization of individual descriptors. At first glance, classical statistical features such as kurtosis and skewness appear to be ranked lower than spectral entropy and composite interaction features. However, closer examination indicates strong conceptual agreement regarding the dominant signal characteristics governing partial discharge (PD) classification. This finding is important because it demonstrates that different feature selection mechanisms may emphasize distinct mathematical representations of the same underlying physical phenomena. Consequently, the observed differences should be interpreted not as inconsistency, but as evidence that PD-related information is encoded redundantly across multiple feature domains (Yao et al., 2020; Du et al., 2023).

Previous studies on partial discharge (PD) signal classification have predominantly relied on handcrafted feature sets derived from time-domain statistics, frequency-domain summaries, and phase-resolved partial discharge (PRPD) representations. In particular, time-domain statistical descriptors such as kurtosis, skewness, and crest factor are frequently reported as dominant features due to their sensitivity to impulsive discharge events and asymmetric amplitude distributions. Frequency-domain features, including FFT-based energy measures and spectral peaks, are also commonly employed to characterize discharge-related spectral content, while PRPD-based quadrant statistics are widely used to encode phase-locked discharge behaviour. In most prior works, these features are evaluated either individually or within relatively small, manually curated feature sets, with limited use of interaction or composite descriptors (Auni et al., 2020; Soh et al., 2022).

Under the Featurewiz–XGBoost framework, spectral entropy and related frequency-domain interaction features dominate the ranking, while classical statistical descriptors appear with lower relative importance. This outcome reflects the sensitivity of ANOVA-based ranking to features that maximize class separability in the spectral domain. The prominence of spectral entropy indicates that PD signals exhibit structured, non-random frequency distributions that are distinct from background noise, confirming that discharge activity introduces measurable disorder and localized energy concentration in the frequency spectrum (L. Sun & Song, 2020; Romphuchaiyapruet & Wattanawongpitak, 2025). This result shows that PD events are not solely characterized by impulsiveness in the time domain, but also by organized patterns in spectral space.

In contrast, the MLJAR internal feature selection framework prioritizes composite and interaction features that combine spectral entropy with fractal dimension, RMS, variance, and PRPD-related measures. The dominance of these hybrid descriptors indicates that PD classification benefits from capturing joint relationships between signal complexity, amplitude modulation, and phase-resolved behaviour. This suggests that PD signals exhibit multi-scale structure, where impulsiveness, complexity, and energy distribution interact in a non-linear manner. Importantly, many of the highest-ranked

MLJAR features implicitly encode impulsiveness and asymmetry through composite formulations, even when kurtosis or skewness do not appear explicitly among the top-ranked variables.

When compared with previous studies, the apparent reduction in ranking of classical time-domain statistics can be explained primarily by differences in feature construction and selection methodology rather than by disagreement in physical interpretation. Earlier works typically evaluate kurtosis, skewness, and crest factor as standalone descriptors within limited feature spaces, causing these features to emerge as dominant indicators of PD activity. In contrast, the present study employs automated feature engineering and independent, interaction-aware selection frameworks, which redistribute importance toward composite features that subsume impulsiveness and asymmetry within higher-order representations. As a result, classical statistical features may appear ranked lower, not because they are less informative, but because their diagnostic contribution is absorbed into interaction features that capture the same physical properties more comprehensively.

Overall, this finding refines rather than contradicts existing literature. It demonstrates that the perceived importance of individual PD features is strongly influenced by the feature selection paradigm. When interaction-aware and model-agnostic selection is applied, PD discrimination is driven not by isolated descriptors, but by coupled representations of complexity, spectral organization, and impulsiveness. Thus, the present results extend previous studies by showing that impulsive and asymmetric discharge behaviour remains central to PD characterization, whether expressed explicitly through traditional time-domain statistics or implicitly through composite spectral and structural features.

5.3 Strengths of the Proposed Approach

A principal strength of this study lies in the utilization of long-term, multi-site field monitoring data acquired from operational European overhead power lines, which directly supports realism and deployment relevance in partial discharge (PD) analytics (Florkowski, 2024; Sikorski & Gielniak, 2025). Unlike laboratory-based datasets, the use of field-acquired measurements enables the evaluation of model performance under authentic operating conditions, capturing signal variability arising from environmental interference, load dynamics, and insulation aging processes (Orellana et al., 2023; Florkowski, 2024). The multi-year temporal coverage and geographic diversity of the dataset further allow for the assessment of model stability and consistency across heterogeneous real-world conditions (Seitz et al., 2024; Sikorski & Gielniak, 2025).

Another key strength is the adoption of a comprehensive multi-domain feature engineering framework encompassing time-domain statistics, frequency-domain spectral descriptors, time–frequency representations, and phase-resolved partial discharge (PRPD) features (Aldosari et al., 2023; Long et al., 2024). This design explicitly acknowledges that different PD mechanisms manifest complementary characteristics across multiple signal domains, thereby improving class separability compared to single-domain feature representations (Aldosari et al., 2023; Long et al., 2024). By integrating physically meaningful descriptors linked to discharge energy, temporal behavior, spectral content, and phase dependency, the proposed feature set provides a robust and interpretable representation of PD phenomena (Florkowski, 2024; Long et al., 2024).

The use of two independent feature selection pipelines, namely Featurewiz and MLJAR, constitutes an additional methodological strength by enabling cross-validation of feature relevance under distinct selection philosophies (Seitz et al., 2024; C. Zhang et al., 2024). The substantial overlap observed between the features selected by both methods provides internal evidence of selection stability and reduces the risk of algorithm-specific bias in high-dimensional feature spaces (Seitz et al., 2024; C. Zhang et al., 2024). This convergence strengthens confidence that the retained features capture

physically meaningful and diagnostically relevant characteristics rather than dataset-specific artifacts (Long et al., 2024; Seitz et al., 2024).

Furthermore, the evaluation framework employs multiple complementary performance metrics, including accuracy, precision, recall, F1-score, and ROC-AUC, enabling a risk-aware interpretation of model behavior under class imbalance (Zemouri & Levesque, 2023; Adhikari et al., 2024). Such comprehensive reporting supports application-driven assessment, where different operational contexts may prioritize fault sensitivity, false-alarm reduction, or overall classification robustness (Zemouri & Levesque, 2023; Adhikari et al., 2024). By pairing quantitative performance metrics with uncertainty estimates derived from stratified cross-validation, the proposed approach facilitates informed decision-making for industrial PD monitoring deployments (Florkowski, 2024; Seitz et al., 2024).

CHAPTER 6

CONCLUSION

6.1 Summary and Key Findings

This thesis investigated the classification of high-frequency partial discharge (PD) signals from medium-voltage overhead lines using traditional machine learning models under two independent feature selection regimes: Featurewiz and MLJAR. The study demonstrated that, despite substantial differences in feature construction, dimensionality, and selection strategy, both regimes consistently identified a stable core of informative descriptors spanning time-domain statistics, frequency-domain summaries, complexity measures, and phase-resolved patterns.

Comparative evaluation of machine learning models showed that ensemble-based and nonlinear classifiers achieved the strongest overall performance. Under the Featurewiz feature selection track, the Random Forest classifier attained the highest classification accuracy of 91.91% with a corresponding F1-score of 71.84%, demonstrating competitive performance using a compact and interpretable feature set. In contrast, the MLJAR-based approach achieved its best performance using a Random Forest classifier with an accuracy of 91.83%, precision of 81.50%, recall of 68.94%, F1-score of 73.11%, and ROC-AUC of 90.63%.

While the MLJAR-based framework yielded a modest improvement in F1-score, this improvement was accompanied by a substantial increase in feature dimensionality, highlighting an important trade-off between classification performance and model complexity. The Featurewiz-based approach, by comparison, achieved near-equivalent accuracy

with a significantly reduced feature set, offering advantages in computational efficiency and interpretability that are desirable for practical deployment.

Feature importance analysis further revealed that classical statistical indicators of impulsiveness and asymmetry remain diagnostically relevant, even when their contribution is implicitly embedded within higher-order interaction features. This observation clarifies that apparent differences in feature rankings across selection frameworks arise primarily from methodological representation rather than disagreement in the underlying physical characteristics of partial discharge phenomena.

Overall, the results confirm that robust PD classification can be achieved through multi-domain feature integration and model-agnostic feature selection strategies. The findings presented in this thesis provide a physically grounded and practically viable basis for the development of reliable partial discharge diagnostic systems suitable for real-world monitoring environments.

6.2 Limitations of the Study

A key limitation of this study arises from the exclusive reliance on long-term, multi-site field monitoring data acquired from operational overhead power lines, which are subject to inherently uncontrolled measurement conditions (Florkowski, 2024; Sikorski & Gielniak, 2025). Field-acquired partial discharge (PD) signals are therefore unavoidably affected by environmental noise sources, including electromagnetic interference, weather-induced disturbances, and load-dependent fluctuations, which can obscure weak discharge signatures and reduce classification reliability (Orellana et al., 2023; Florkowski, 2024). Further variability introduced by insulation aging, material heterogeneity, and temporal degradation processes complicates the separation of discharge-specific features from background interference (Seitz et al., 2024; Sikorski & Gielniak, 2025). Although such real-world conditions enhance ecological validity and reflect prac-

tical deployment scenarios, they limit the isolation of fundamental PD mechanisms under high signal-to-noise clarity (Florkowski, 2024; Long et al., 2024).

The absence of complementary experimental data acquired under controlled laboratory conditions restricts the evaluation of intrinsic feature sensitivity and noise-independent discriminative capability (Aldosari et al., 2023; Long et al., 2024). Laboratory-based PD measurements, characterized by well-defined defect geometries, stable operating conditions, and high signal-to-noise ratios, provide an essential reference for validating feature relevance under idealized conditions (Aldosari et al., 2023; Long et al., 2024). As a result, the generalizability of the proposed framework across the full spectrum of PD operating conditions cannot be fully established using field data alone (Florkowski, 2024; Seitz et al., 2024). Future work should therefore integrate controlled experimental datasets with real-world monitoring data to enable systematic benchmarking between noise-free and operational environments and to strengthen the physical interpretability of PD classification models (Long et al., 2024; Sikorski & Gielniak, 2025).

A further limitation concerns the geographic and climatic context of the training data (Florkowski, 2024; Sikorski & Gielniak, 2025). The dataset utilized in this study was acquired from monitoring stations located in Czechia and Slovakia, regions characterized by temperate continental climates with distinct four-season annual cycles (Florkowski, 2024; Sikorski & Gielniak, 2025). Such climatic variability introduces environmental conditions including temperature fluctuations, humidity patterns, precipitation regimes, and seasonal insulation stress variations that may not be representative of regions with different climate characteristics (Orellana et al., 2023; Azam et al., 2024).

Specifically, four-season climate dynamics may influence PD activity through temperature-dependent insulation behavior, humidity-driven surface contamination effects, and seasonally varying electromagnetic interference patterns (Orellana et al., 2023; Florkowski, 2024). The transferability of models trained on this data to regions with fundamentally different climate regimes, such as tropical or equatorial environments with only wet and dry seasons, remains unvalidated (Azam et al., 2024; Florkowski, 2024). For example,

deployment in Malaysia may encounter distinct PD signal characteristics, environmental noise profiles, and insulation stress mechanisms that could affect model performance (Azam et al., 2024; Florkowski, 2024). This limitation underscores the importance of region-specific validation and potential domain adaptation strategies when deploying PD classification models across diverse geographic and climatic contexts (Friebe & Jenau, 2024; Seitz et al., 2024).

The presence of precision–recall performance gaps, where even the best-performing models achieved recall values ranging from 67.09% to 72.32%, indicates that a proportion of fault instances remain undetected despite improved sensitivity to the minority class (Aldosari et al., 2023; Zemouri & Levesque, 2023). This constraint motivates the adoption of threshold calibration, cost-sensitive learning strategies, and class rebalancing techniques in scenarios where false-negative risk is operationally critical (Zemouri & Levesque, 2023; Adhikari et al., 2024).

6.3 Transition to Future Work

External validation across additional assets, monitoring sites, and acquisition modalities constitutes a primary requirement for establishing the broader transferability and generalizability of the developed PD classification models beyond the present dataset (Seitz et al., 2024; Sikorski & Gielniak, 2025). Such validation should explicitly include datasets collected from diverse geographic regions and climatic regimes to assess the impact of environmental variability on model performance, directly addressing the identified limitation associated with four-season versus two-season climate contexts (Azam et al., 2024; Florkowski, 2024).

Differences in noise statistics, electromagnetic coupling paths, sensor configurations, and environmental interference patterns across deployment contexts further motivate the investigation of domain adaptation and robustness-oriented learning strategies

(Florkowski, 2024; Friebe & Jenau, 2024). Techniques such as transfer learning, domain adversarial training, and environment-aware data augmentation could be explored to enhance model robustness under heterogeneous operational and climatic conditions (Azam et al., 2024; Seitz et al., 2024).

In parallel, future work should focus on explicitly addressing class imbalance and asymmetric operational risk through calibrated decision thresholds, cost-sensitive learning objectives, and advanced resampling strategies to improve minority fault detection sensitivity (Zemouri & Levesque, 2023; Adhikari et al., 2024). Additionally, data augmentation and advanced denoising approaches are widely recognized as practical mechanisms for improving robustness to interference and enhancing generalization when operating on noisy field-acquired PD signals (X. Bao et al., 2024; Friebe & Jenau, 2024).

Finally, principled feature set reduction combined with interpretability analysis and uncertainty quantification should be pursued to support deployable monitoring workflows that balance classification performance, computational efficiency, and engineering trust (Vera et al., 2023; Florkowski, 2024). The integration of explainable artificial intelligence techniques with the developed models would further enhance their suitability for industrial deployment scenarios where transparent diagnostics and confidence-aware decision-making are critical for asset maintenance planning (Vera et al., 2023; Seitz et al., 2024).

Beyond the evaluated classifiers, future studies may explore a broader range of machine learning and deep learning architectures, including ensemble hybrids, attention-based neural networks, and transformer-inspired models, to capture more complex temporal and spectral dependencies in PD signals (Zemouri & Levesque, 2023; Seitz et al., 2024). Such models have demonstrated improved representation learning capabilities in noisy and non-stationary signal environments, making them promising candidates for real-world PD monitoring applications (Florkowski, 2024; Sikorski & Gielniak, 2025).

In addition, the adoption of advanced denoising techniques, such as wavelet-domain

adaptive filtering, empirical mode decomposition variants, and learning-based denoising autoencoders, could further improve signal quality prior to feature extraction and classification (X. Bao et al., 2024; Friebe & Jenau, 2024). These approaches enable the suppression of non-stationary interference while preserving discharge-relevant signal components, which is critical for enhancing classification robustness under field-acquired noise conditions (Orellana et al., 2023; Florkowski, 2024).

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